Power Plant (Mod 6/2012) – General Data

1. General

Each power plant consists of a United Aircraft PT6A–34 engine, engine mount assembly, exhaust stubs, and the equipment and accessories required for the supply of fuel and air for the correct functioning of the engine under all operating conditions. The engine is supported at three points by means of vibration isolators attached to the front face of the engine nacelle structure. Enclosing the engine are two removable engine cowlings, the lower cowling housing a snow and ice deflector system. The power plant is protected by fire detection and fire extinguishing systems, and two fireseals fitted to the front and rear of the engine compressor intake.

2. Troubleshooting

TROUBLE	POSSIBLE CAUSE	CORRECTION
Failure to light up.	Ignition system defective.	Refer to PT6A Engine Maintenance Manual, Ignition Section. 74-00-00, Ignition – General Data
Acceleration rates varying between engines.	Incorrect engine rigging.	Recheck engine rigging and adjust as required.
Propeller rpm and torque pressure fluctuation.	Interplay between propeller constant speed and overspeed governors.	Check CSU governor for correct operation.

Note For further information on power plant troubleshooting, refer to PT6A Engine Maintenance Manual or 72-00-00, Engine – General Data.

3. Removal

Refer to Figure 1 for details.

A. Remove Power Plant

1) With electrical power ON, close fuel emergency shut-off valve using firewall shut-off valve switch on emergency panel. Close fuel shut-off valve by setting fuel lever to OFF position.

2) Disconnect electrical power.

3) Support rear fuselage using steady jacks and bar.

4) Disconnect leads from upper nacelle fire detector.

5) Remove cowlings (refer to 71-10-00, Cowlings – General Data), and access panel from top of nacelle.

6) Remove propeller (refer to 61-10-00, Propeller Assembly – General Data).

7) Disconnect power, propeller, and fuel shut-off controls at engine.

8) Disconnect main fuel feed, and fuel purge lines at main firewall.

9) Disconnect electrical connectors and leads at lower left firewall.

10) Remove electrical leads from starter/generator.

11) Disconnect oil breather line at main firewall.

12) Disconnect bleed air line at main firewall.

13) Attach engine sling SD12541 and take up power plant weight (refer to 07-00-00, Lifting and Shoring – General Data).

14) Remove single attachment bolt, using wrench SD12524, from each vibration isolator and lower power plant (refer to 71-20-11, Vibration Isolator – Maintenance Practices).

15) Remove vibration isolators from each side of engine easing (refer to 71-20-11, Vibration Isolator – Maintenance Practices).

16) Lower power plant onto maintenance stand. Remove engine sling and upper vibration isolator (refer to 71-20-11, Vibration Isolator – Maintenance Practices).

B. Remove Power Plant Accessories

CAUTION BLANK OFF COMPRESSOR INTAKE PRIOR TO REMOVAL OF ACCESSORIES.

1) Remove fuel and oil drain lines, and drain line brackets.

2) Remove oil cooler (refer to 79-20-12, Oil Cooler (Mod 6/1438 and Mod 6/1481) – Maintenance Practices).

3) Remove from engine accessory case the oil pressure switch, pressure line, switch mounting bracket, oil temperature bulb and adapter and oil pressure transmitter from mounting face (refer to 79-00-00, Oil – General Data).

4) Remove external oil lines from engine accessory case.

5) Remove fire detectors.

6) Remove the following controls from the engine (refer to 76-00-00, Engine Controls – General Data):

   a) Propeller control complete with brackets and hardware for propeller governor.

   b) Beta control cam, cam follower lever, and FCU actuating lever complete with hardware from accessory gear case.

   c) Fuel shut-off link lever bracket on top of FCU.

7) Remove starter/generator cooling duct (refer to 80-10-11, Starter/Generator – General Data).

8) Remove bleed air line, and intake deflector line.

9) Remove starter/generator and drive mount pad and clamp (refer to 80-10-11, Starter/Generator – General Data).

10) Remove exhaust nozzles (refer to 78-00-00, Exhaust – General Data).

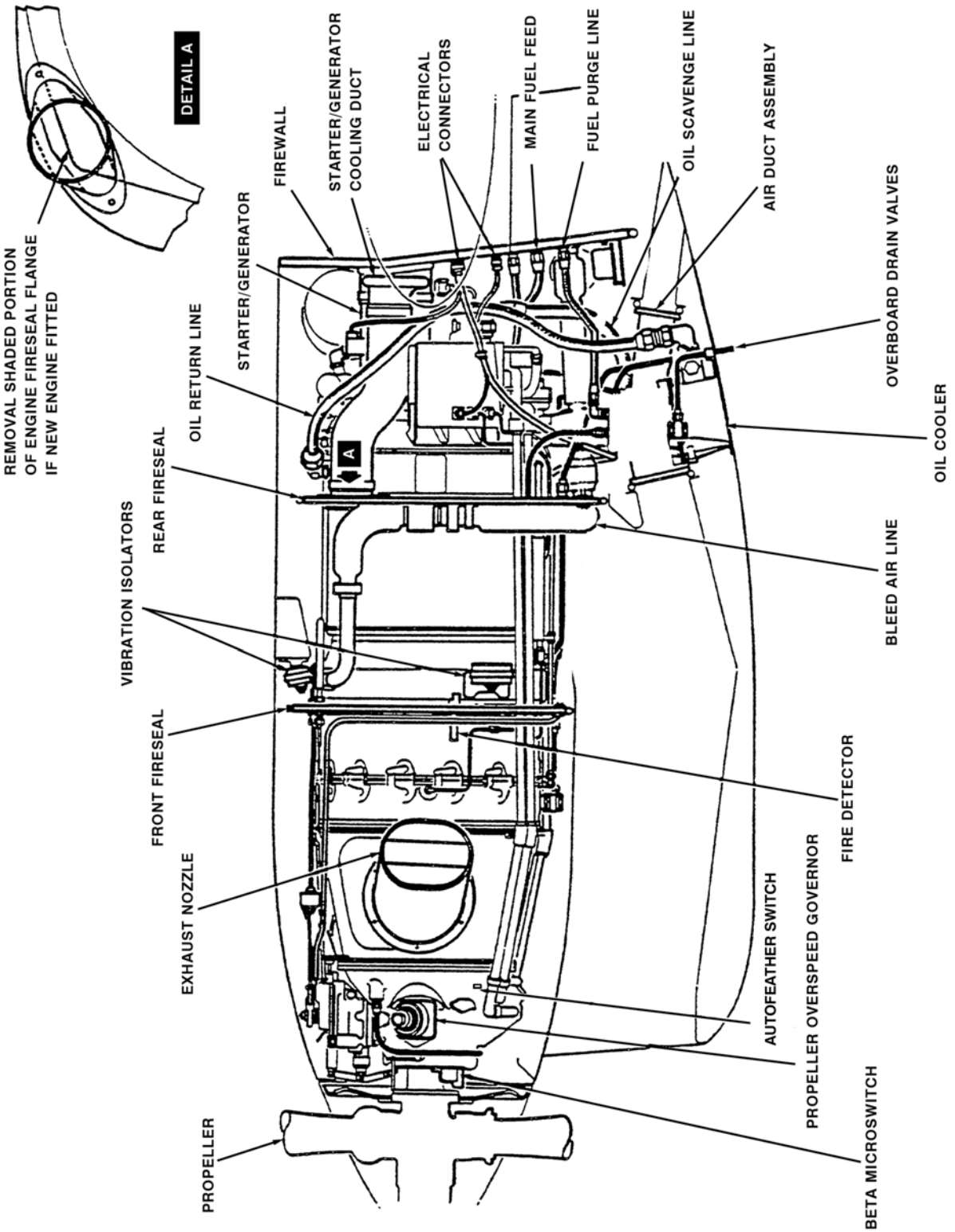
11) Remove electrical harness and leads (refer to 71-50-00, Electrical Harness – General Data).

12) From reduction gear case, remove torque pressure transmitter and vent line, autofeather switch and hose, beta microswitch and bracket (refer to 61-24-13, Beta Microswitch Assembly (Mod 6/2022) – General Data), propeller overspeed governor, (refer to 61-20-21, Overspeed Governor – Maintenance Practices), and propeller tachometer generator.

13) Remove gas generator tachometer generator from engine accessory case.

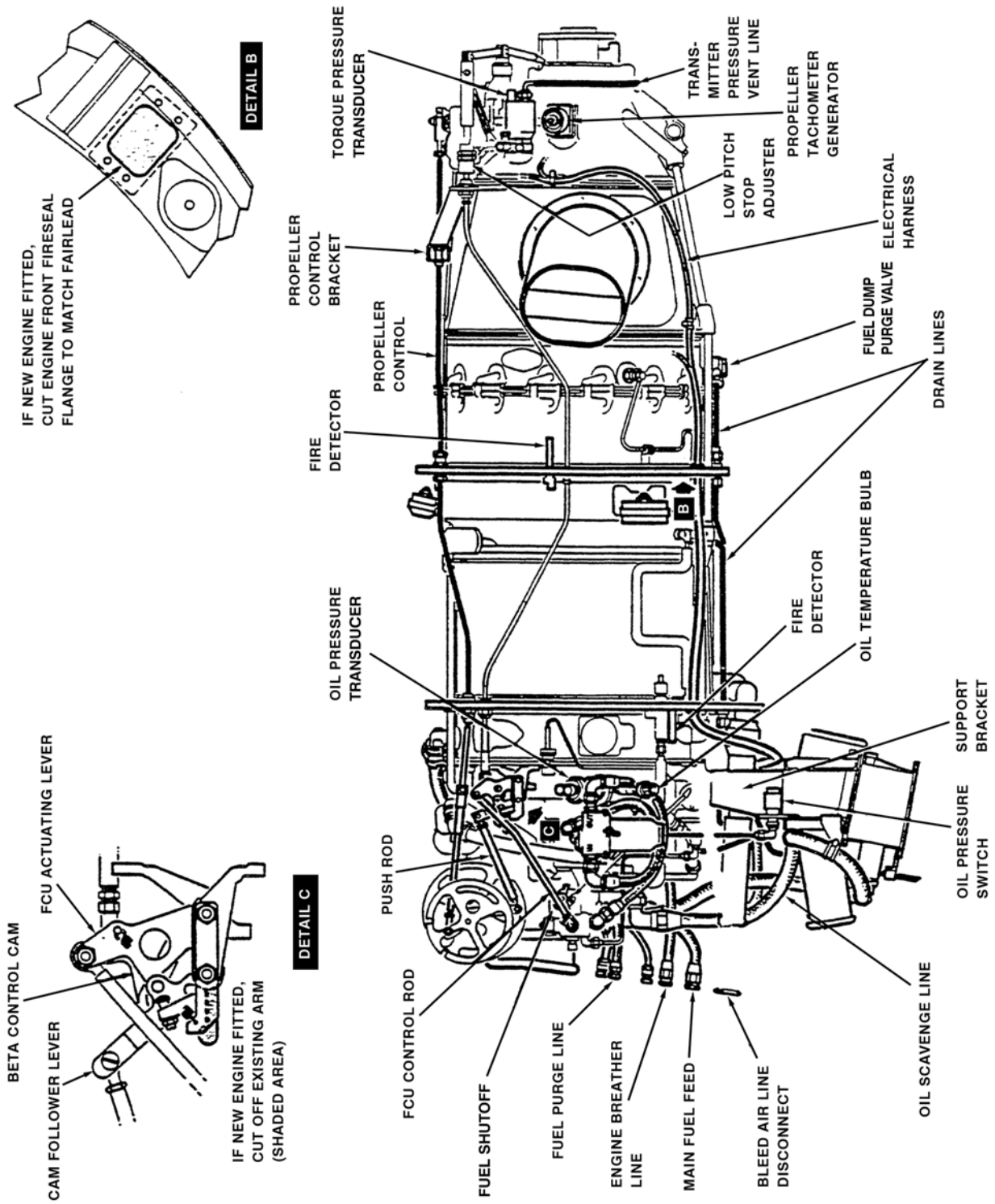
14) Remove front and rear fireseals.

15) Remove air duct assembly and support bracket from underside of accessory gearbox.



Power Plant Installation
Figure 1 (Sheet 1 of 2)

Print Date: 2025-09-18



Power Plant Installation
Figure 1 (Sheet 2 of 2)

Print Date: 2025-09-18

4. Installation

CAUTION **IF PREVIOUSLY INSTALLED ENGINE HAS BEEN REMOVED BECAUSE OF METAL CONTAMINATION OF OIL SYSTEM, PROPELLER, PROPELLER OVERSPEED GOVERNOR, OIL COOLER, AND ASSOCIATED PIPE LINES AND HOSES MUST BE THOROUGHLY CLEANED BY FLUSHING. IF OIL COOLER CANNOT BE ADEQUATELY CLEANED BY FLUSHING, IT MUST BE REPLACED.**

Note Unless otherwise specified, install engine nuts and/or bolts to torque values stated in PT6A Engine Maintenance Manual.

Refer to Figure 1 for details.

A. Install Power Plant Accessories

CAUTION **BLANK OFF COMPRESSOR INTAKE PRIOR TO INSTALLATION OF ACCESSORIES.**

Note All splines of accessory drives to be lubricated with Molykote M77 or Plastilube No. 3 (Warren Refining Co., Cleveland, Ohio).

1) Install support bracket to underside of accessory gearbox.

2) Install air duct assembly to bracket.

3) Install front and rear fireseals.

Note If new engine fitted, cut engine front fireseal flange to match front fireseal electrical harness fairlead and rear fireseal flange to match starter/generator cooling duct intake (refer to Figure 1 Detail A and B).

4) Install gas generator tachometer generator to engine accessory case.

5) Install to reduction gear case, the torque pressure transmitter and pressure line, autofeather switch and hose, beta microswitch and bracket (refer to 61-24-13, Beta Microswitch Assembly (Mod 6/2022) – General Data), propeller overspeed governor and drive (refer to 61-20-21, Overspeed Governor – Maintenance Practices), and propeller tachometer generator.

6) Install electrical harness and leads (refer to 71-50-00, Electrical Harness – General Data).

7) Install exhaust nozzles (refer to 78-00-00, Exhaust – General Data).

8) Install gasket, drive mount pad (with dowel at 12 o'clock position), and starter/generator (refer to 80-10-11, Starter/Generator – General Data).

9) Install bleed air line, and intake deflector line. Torque each Janitrol clamp to value quoted on clamp body.

10) Install starter/generator cooling duct (refer to 80-10-11, Starter/Generator – General Data).

11) Install the following controls to the engine (refer to 76-00-00, Engine Controls – General Data):

a) Beta control cam, cam follower lever, and FCU actuating lever complete with hardware.

b) Propeller control complete with brackets and hardware for propeller governor.

c) Fuel shut-off link lever bracket to the top of the FCU.

12) Install fire detectors.

13) Install external fuel and oil lines to engine accessory case (refer to 79-00-00, Oil – General Data).

14) Install to engine accessory case, the oil pressure switch mounting bracket, oil pressure switch and line, oil temperature bulb and adapter, and oil pressure transmitter.

15) Install oil cooler (refer to 79-20-12, Oil Cooler (Mod 6/1438 and Mod 6/1481) – Maintenance Practices).

16) Install fuel and oil drain lines, and drain lines bracket to oil cooler.

17) Rig engine controls (refer to 76-00-00, Engine Controls – General Data and 76-10-00, Power Control – Maintenance Practices).

B. Install Power Plant

1) Install engine bracket assembly and upper vibration isolator on engine casing using the four attachment bolts (refer to 71-20-11, Vibration Isolator – Maintenance Practices).

2) Install engine sling SD12541 and take up power plant weight (refer to 07-00-00, Lifting and Shoring – General Data).

3) Remove power plant from maintenance stand.

4) Install vibration isolators on each side of engine casing using the four attachment bolts per isolator on each side of engine casing (refer to 71-20-11, Vibration Isolator – Maintenance Practices).

5) Raise power plant into position in engine nacelle.

6) Secure power plant to nacelle with vibration isolators attachment bolts (refer to 71-20-11, Vibration Isolator – Maintenance Practices). Remove engine sling.

7) Connect bleed air line to firewall connection.

8) Connect oil breather line to firewall connection.

9) Connect electrical leads to starter/generator.

10) Connect electrical connectors and leads at lower left main firewall.

11) Connect power, propeller, and fuel shut-off controls to engine.

12) Install propeller (refer to 61-10-00, Propeller Assembly – General Data).

13) Install cowlings (refer to 71-10-00, Cowlings – General Data).

14) Connect leads to upper nacelle fire detector.

15) Connect electrical power.

16) Rig power plant controls (refer to 76-00-00, Engine Controls – General Data to 76-10-11, Overhead Console – General Data).

Note Ensure low pitch stop adjuster spring has been removed.

17) Install access panel in top of nacelle.

18) Remove rear fuselage steady jacks and bar.

19) Check for fuel and oil leaks on subsequent engine run.

20) Carry out post installation procedures, refer to Para 10.

5. Ground Running – General Precautions

   A. Action Prior to Ground Running

      1) Position aircraft facing into wind.

      2) Chock wheels.

      3) Fire extinguisher in position.

      4) Ensure area around aircraft is free of loose objects.

      5) Check engine oil level (refer to 12-10-25, Oil System – General Data).

      6) Check engine intakes and exhausts are free from foreign objects.

      7) Ensure danger areas around aircraft are clear of personnel.

6. Engine Depreservation

   A. Depreserve Engine

      1) Fill engine oil tank (refer to 12-10-25, Oil System – General Data).

         Note Oil system does not require any depreservation unless different oil is to be used in system. If different oil used, flush system after Engine Leak Test Run (refer to 12-10-25, Oil System – General Data).

      2) Carry out preliminary action prior to ground running (refer to Para 5.).

      3) Pull IGNITION circuit breakers and fit safety clip.

      4) POWER SOURCE switch – EXTERNAL (Pre Mod 6/2314); or EXTERNAL switch – ON (Post Mod 6/2314).

      5) Booster pumps – FWD BOOST or AFT BOOST for right or left engine.

      6) Disconnect inlet fuel line at bottom fuel purge/dump valve adapter (refer to PT6A Engine Maintenance Manual); to permit drainage into container.

      7) Ensure external electrical power is connected and MASTER switch is ON (Pre Mod 6/2314); or EXTERNAL switch is ON (Post Mod 6/2314)

      8) Power lever – IDLE.

      9) Fuel lever – ON.

      10) START switch – hold at LEFT or RIGHT (as applicable) for approximately 10 seconds, then release switch.

      11) Check fuel flow from line into containers is free of preservative fluid.

      12) Fuel lever – OFF.

      13) Check fuel flow stops flowing release start switch lever.

CAUTION **OBSERVE THE STARTER OPERATING TIME LIMITS AS FOLLOWS:**

- **25 SECONDS ON, 1 MINUTE OFF**
- **25 SECONDS ON, 1 MINUTE OFF**
- **25 SECONDS ON FOLLOWED BY A 30 MINUTE COOLING-OFF PERIOD**

14) Fuel booster pump – OFF.

15) Reconnect fuel line to bottom of fuel dump and purge valve.

Note An eight hour soaking period is recommended after fuel system deprivations to allow bypass valve diaphragm to return to the required degree of pliability, and also to permit synthetic packing to swell and provide satisfactory seals.

16) Check for leaks in engine fuel system.

17) Check safety and security of all connections.

7. Fuel Drains and Dump Test (Wet Run)

A. Test FCU Purge Valve

1) Remove the fuel manifold purge line aft of the firewall at the bulkhead fitting. Position container to receive spillage.

2) Pull applicable IGNITION L or IGNITION R circuit breaker.

3) Power lever – IDLE.

4) Booster pump on – FWD BOOST or AFT BOOST (as applicable).

5) Engine fuel lever – OFF.

6) START switch – hold to LEFT or RIGHT (as applicable) to motor engine.

CAUTION OBSERVE THE STARTER OPERATING TIME LIMITS AS FOLLOWS:

- **25 SECONDS ON, 1 MINUTE OFF**
- **25 SECONDS ON, 1 MINUTE OFF**
- **25 SECONDS ON FOLLOWED BY A 30 MINUTE COOLING-OFF PERIOD**

7) Observe fuel flows from FCU purge valve fitting.

8) Move engine fuel lever to OFF.

9) Observe fuel flows from fuel manifold purge line at the bulkhead fitting.

10) Release starter switch and observe fuel flows from combustion chamber drain lines.

11) Check fuel pump and generator seal overboard drains for leakage.

12) Switch off booster pumps.

13) Reinstall the fuel lines C6PF1026–61 LH and C6PF1026–59 RH.

WARNING ENSURE RESIDUAL FUEL IN THE ENGINE COMBUSTION CHAMBER HAS STOPPED DRAINING BEFORE ATTEMPTING TO CARRY OUT ENGINE START.

Table 1 Operating Limits

OPERATING CONDITION	PT6A–34 OPERATING LIMITS					
POWER SETTING	TORQUE PSI (Ti)	TURBINE TEMP °C (T_5)	GAS GEN SPEED % (N_G)	PROP SPEED % (N_P)	OIL PRESS PSI (L_P)	OIL TEMP °C (LT)
Starting		1090 max (limit 2 secs)				–40 min
Idle		685 max	52		40 min	–40 to 99
Acceleration	68.7 max (limit 2 secs)	825 max (limit 2 secs)	102.6 max	110 max	85 to 105	0 to 99
Take-off and max continuous	50 max	790 max	101.6	96	85 to 105	10 to 99

CAUTION IF OIL PRESSURE DROPS BELOW 85 PSI WHEN GAS GENERATOR SPEED IS GREATER THAN 72% N_G, ENGINE RUN SHOULD BE TERMINATED.

8. Engine Starting and Shutdown

A. Prestart Checks

Note All switches to be off, unless otherwise stated.

1) Parking brake – ON (check pressure).

2) Circuit breakers – IN.

3) MASTER switch – ON (Pre Mod 6/2314).

4) POWER SOURCE switch – As required (Pre Mod 6/2314); or EXTERNAL and/or BATTERY switch – ON (Post Mod 6/2314).

5) IGNITION switch – NORMAL.

6) GENERATORS switches – OFF. Check caution message displayed.

7) BUS TIE switch – NORMAL.

8) CAUTION LIGHTS test switch – TEST.

9) Check position of blades on each propeller and set propeller levers and power levers accordingly as follows:

a) Propeller latches (if fitted) engaged.

CAUTION WITH ENGINES STOPPED AND PROPELLER BLADE LATCHES ENGAGED, NO ATTEMPT MUST BE MADE TO MOVE POWER LEVERS FORWARD TO IDLE OR MORE THAN 1.25 INCHES AFT OF IDLE.

1 Propeller levers – FEATHER (full INCREASE for pressure altitudes below 3000 feet when temperature is below 50°F (10°C) OAT).

2 Power levers – Zero thrust reference lines aligned.

CAUTION **OBSERVE OPERATING LIMITATIONS (REFER TO Table 1 — Operating Limits) DURING ENGINE RUNNING.**

b) Propeller feathered.

1 Propeller levers – FEATHER (full INCREASE for pressure altitudes below 3000 feet when temperature is below 50°F (10°C) OAT).

2 Power levers – IDLE.

CAUTION **WITH ENGINES STOPPED DO NOT ATTEMPT TO MOVE POWER LEVERS AFT OF IDLE TOWARD THE ZERO THRUST STOP/START LINE OR PROPELLER CONTROL LINKAGE MAY BE DAMAGED.**

10) Fuel levers – OFF (fully aft).

11) FIRE WARNING test switch – test.

12) FUEL EMERGENCY shut-off switches – NORMAL.

13) FUEL SELECTOR – NORMAL.

14) BOOST PUMP switches – FWD BOOST and AFT BOOST. (Check all four boost pump indication are not displayed.)

B. Engine Starting, Using External Electrical Power

CAUTION **IN AMBIENT CONDITIONS OF 90°F (32°C) AND OVER, GROUND OPERATION OF ENGINES IN FEATHERED, DISCING (ZERO THRUST), OR REVERSE, MUST BE KEPT TO AN ABSOLUTE MINIMUM (LESS THAN ONE MINUTE), OR INTERNAL NACELLE TEMPERATURES WILL EXCEED SAFE LIMITS.**

1) Carry out prestart checks as in Para 8.A.

2) AVIONICS START BATTERY switch (Post Mod 6/2315) – AUTO.

WARNING **ON AIRCRAFT WITH THE GPU RECEPTACLE LOCATED FORWARD OF THE PROPELLER (S.O.O. 6371), THE GPU PLUG SHOULD BE DISCONNECTED BEFORE STARTING THE LEFT ENGINE. DISCONNECTING THE GPU PLUG AFTER STARTING THE LEFT ENGINE PUTS THE GROUND CREW AT RISK OF BEING STRUCK BY THE PROPELLER.**

3) START switch – Hold at LEFT or RIGHT, as required. Check for oil pressure rise. Check appropriate L or R GENERATOR caution light goes out.

4) Allow gas generator speed to stabilize above 12% N_G, then move applicable engine fuel lever ON.

CAUTION **DO NOT SELECT ENGINE FUEL LEVER ON (ALLOWING FUEL TO FLOW) BEFORE STABILIZED RPM ARE REACHED. THE MINIMUM SPEED TO OBTAIN SATISFACTORY LIGHT-UP IS 12% N_G. ANY START ATTEMPT BELOW THIS SPEED IS CONSIDERED HAZARDOUS AND SHOULD BE MONITORED CLOSELY TO PREVENT AN OVER TEMPERATURE CONDITION.**

5) Light-up – Check that engine accelerates to normal idle rpm for prevailing ambients and turbine temperature does not exceed 1090°C. (Time limit at 1090°C is two seconds; operators are advised to keep hand on engine fuel lever during this period).

CAUTION **WHEN A GAS GENERATOR FAILS TO LIGHT UP WITHIN 10 SECONDS OF ADVANCING ENGINE FUEL LEVER, OR ENGINE OVER TEMPERATURE IS IMMINENT, SET ENGINE FUEL LEVER OFF TO STOP FUEL FLOW, AND CONTINUE MOTORING THE ENGINE WITH STARTER FOR APPROXIMATELY 10 SECONDS. OBSERVE STARTER LIMITATIONS (REFER TO Para 8.E.5)) AND ALLOW THE ENGINE TO COME TO A COMPLETE STOP BEFORE ATTEMPTING A NEW START.**

6) START switch – Release. Check appropriate L or R GENERATOR light comes on.

CAUTION **IF EITHER L OR R GENERATOR CAUTION LIGHT DOES NOT COME ON, IMMEDIATELY SELECT BUS TIE SWITCH TO OPEN AND EXTERNAL/BATTERY SWITCH TO OFF. SHUTDOWN ENGINE(S) AND RECTIFY UNSERVICEABILITY.**

7) Oil pressure – At least 40 psi, and ENGINE OIL PRESSURE caution light out.

8) Repeat Para 8.B.3) through Para 8.B.7) to start second engine.

9) Propeller levers – Full INCREASE.

10) Power levers – Advance to idle N_G + 15%.

CAUTION **IF THERE IS RESISTANCE TO FORWARD MOVEMENT OF THE POWER LEVERS, INDICATING FAILURE OF LATCHES TO WITHDRAW, THE POWER LEVERS MUST NOT BE FORCED AHEAD OR DAMAGE TO CONTROL MECHANISM MAY BE CAUSED.**

Note If blade latches were engaged when engines were started, a momentary movement of the power levers toward reverse should be made to ensure latch disengagement, before advancing power levers.

11) POWER SOURCE switch – BATTERY (Pre Mod 6/2314); or BATTERY switch – ON (Post Mod 6/2314).

12) GENERATORS switches – Individually RESET and release to ON. Check caution message displayed.

13) Check generator load by selection.

CAUTION **DO NOT SWITCH EITHER GENERATOR TO ON WITHOUT FIRST ADVANCING APPLICABLE POWER LEVER TO IDLE N_G + 15%. SUBSEQUENTLY DO NOT RETARD POWER LEVER(S) FROM THIS SETTING UNTIL GENERATOR LOAD IS 100 AMPS OR LESS. THE GENERATOR(S) MUST BE SWITCHED OFF IF POWER IS REDUCED AND THIS CONDITION IS NOT MET.**

14) Disconnect external electrical power.

C. Engine Starting, Using Battery

1) Carry out prestart checks as in Para 8.A.

2) When a battery start is necessary, start engines as in Para 8.B.3) through Para 8.B.12). If, however, during the start cycle of the first engine there is any reason to doubt the capability of the battery to start the second engine, proceed as follows:

a) Propeller lever of first engine – Full INCREASE.

b) Power lever of first engine – Advance to idle N_G + 15%.

c) GENERATORS switch of first engine – RESET and release to ON. Check that appropriate L or R GENERATOR caution light goes out.

d) Observe battery charge current until load is 40 amperes or less.

e) GENERATORS switch of first engine – OFF.

CAUTION **CROSS GENERATOR STARTING IS DETRIMENTAL TO GENERATOR BRUSHES AND TO REVERSE CURRENT CUT-OUT OPERATION (CONTACT WELDING MAY EVENTUALLY RESULT). CROSS GENERATOR STARTING IS PROHIBITED, HOWEVER, GENERATOR ASSISTED STARTING IS ALLOWED.**

f) Start second engine as in Para 8.B.3) through Para 8.B.7).

g) As second engine speed moves up through 12% N_G, select GENERATORS switch of first engine to RESET and release to ON. Check that appropriate L or R GENERATOR caution light goes out.

h) Complete engine starting procedure as in Para 8.B.9), Para 8.B.10), Para 8.B.12), and Para 8.B.13).

D. Engine Starting of Cold Soaked Engines (Below –20°F (–29°C) OAT), Using Battery

1) Carry out prestart checks as in Para 8.A.

2) Dry motor of first engine to be started (for 5 seconds) by holding START switch at LEFT or RIGHT as required.

3) After approximately one minute, start first engine in accordance with Para 8.B.3) through Para 8.B.7).

4) From stabilized idle with 0°C oil temperature achieved, advance power lever to idle N_G + 15%.

5) GENERATORS switch of normal (operating) engine – RESET and release to ON. Check that appropriate L or R GENERATOR caution light goes out.

6) Observe battery charge current until load is 40 amperes or less.

7) GENERATORS switch of operating engine – OFF.

8) Check L or R GENERATOR caution light illuminates.

CAUTION **CROSS GENERATOR STARTING IS DETRIMENTAL TO GENERATOR BRUSHES AND TO REVERSE CURRENT CUT-OUT OPERATION (CONTACT WELDING MAY EVENTUALLY RESULT). CROSS GENERATOR STARTING IS PROHIBITED, HOWEVER, GENERATOR ASSISTED STARTING IS ALLOWED.**

9) Start second engine as in Para 8.B.3) through Para 8.B.7).

10) As second engine speed moves up through 12° N_G, select GENERATORS switch of first engine to RESET and release to ON. Check that appropriate L or R GENERATOR caution light goes out.

11) Complete engine starting procedure as in Para 8.B.9), Para 8.B.10), Para 8.B.12), and Para 8.B.13).

E. Clearing an Engine – Motor Run

CAUTION **OBSERVE THE STARTER OPERATING TIME LIMITS AS FOLLOWS:**

- **25 SECONDS ON, 1 MINUTE OFF**
- **25 SECONDS ON, 1 MINUTE OFF**
- **25 SECONDS ON FOLLOWED BY A 30 MINUTE COOLING-OFF PERIOD**

Note The following procedure is used to clear an engine when considered necessary to remove internally trapped fuel and vapor, or if there is evidence of a fire within the engine. Air passing through the engine serves to purge fuel, vapor, or fire from the combustion section, gas generator turbine, power turbine, and exhaust system.

1) Engine fuel lever – OFF.

2) Pull applicable IGNITION L or IGNITION R circuit breaker.

3) Power lever – IDLE.

4) Booster pump on – FWD BOOST (RH engine) or AFT BOOST (LH engine) (as applicable).

5) START switch – Hold to LEFT or RIGHT as applicable. Maintain for desired duration, then release switch.

Note If an engine fire is not extinguished during clearing of engine, switch off booster pumps and FUEL EMERGENCY switches. POWER SOURCE switches to OFF. Then vacate aircraft and apply fire extinguisher into exhaust stubs.

F. Engine Shutdown

WARNING IF THERE IS ANY EVIDENCE OF FIRE IN EITHER ENGINE AFTER SHUTDOWN, PROCEED IMMEDIATELY AS FOR CLEARING AN ENGINE IN Para 8.E.

1) Power levers – IDLE.

Note Allow engines to idle for a minimum of one minute at a turbine temperature of 585°C or below. Increase gas generator rpm (N_G) as required to maintain temperature limit.

2) Parking brake – ON.

3) Turbine temperature (T_5) stable.

4) BLEED AIR – OFF.

5) Radios – OFF.

6) GENERATORS – OFF. Check caution message displayed.

CAUTION IF EITHER L OR R GENERATOR CAUTION LIGHT DOES NOT COME ON, IMMEDIATELY SELECT BUS TIE SWITCH TO OPEN AND POWER SOURCE SWITCH TO OFF.

7) Propeller levers – Full INCREASE (latches to be engaged).
or
– FEATHER (latches not to be engaged).

8) Fuel levers – OFF.

Note If propeller blade latches are to be engaged, align zero thrust reference lines prior to, or immediately following, the selection of fuel levers to OFF. The latches will not engage until the propeller rpm has decreased below 30%.

9) BOOST PUMP switches – OFF.

10) AVIONICS START BATTERY switch (Post Mod 6/2315) – OFF.

11) POWER SOURCE switch – As desired.

12) MASTER switch – OFF (Pre Mod 6/2314); or EXTERNAL and BATTERY switches – OFF.

Note During engine shutdown, listen for any unusual scraping or rubbing noises.

9. Engine Leak Test Run

A. Engine Leak Test Run (Cowls Off)

1) Start engine (refer to Para 8.).

2) Check acceleration time from light-up to idle (approximately 12 seconds).

3) Check for:

a) Turbine temperature indication (T_5).

b) Oil pressure indication (LP).

c) Gas generator rpm indication (N_G).

d) Torque pressure indication (Ti).

e) Propeller rpm indication (N_P).

f) Oil temperature indication (LT).

4) With power lever at IDLE, move propeller lever through full range four times, feathering and unfeathering the propeller each time to purge system.

5) Check feathering and unfeathering times at idle setting are approximately 10 and 30 seconds respectively.

CAUTION IF FUEL IS SHUT OFF INADVERTENTLY BY RETARDING FUEL LEVER, DO NOT OPEN FUEL LEVER AGAIN IN ANY ATTEMPT TO REGAIN LIGHT. THE STARTING SEQUENCE MUST BE REPEATED FROM BEGINNING AFTER ALLOWING THE ENGINE TO COME TO COMPLETE STOP.

6) Run at idle speeds until conditions stabilize.

7) Shutdown engine (refer to Para 8.F).

8) Check compressor decelerates freely (minimum time approximately 35 seconds).

9) Check for oil and fuel leaks and security of equipment.

10) Check and top up oil tank (refer to 12-10-25, Oil System – General Data).

11) Install cowlings.

10. Post Installation Procedures

Note The following adjustments and tests are to be carried out after installation of an engine or power plant, but each adjustment or test can be carried out independently if a specific component or setting is to be checked.

A. Adjustments Prior to Ground Running

1) Verify turbine temperature (T_5) indicating system (refer to 77-20-00, Turbine Temperature Indicating System – General Data).

2) Deleted

B. Ground Running Adjustments and Test

Note Unless otherwise stated, engines are to be run with no load applied (generators OFF, bleed air switch OFF, and intake deflector retracted).

Ensure engine to be run is depreserved (refer to Para 6.), prior to carrying out the following adjustments or tests.

1) Carry out engine leak test run (refer to Para 9.A).

2) Adjust gas generator idle speed (N_G) setting.

CAUTION DO NOT DO A CHECK AT BAROMETRIC PRESSURES BELOW 25.2 IN. HG OR ALTITUDES IN EXCESS OF 3500 FEET.

a) Set aircraft altimeter window scale to 29.92. Use the BARO knobs on the pilot's and co-pilot's primary flight display controller to set the altimeter. Record pressure altitude.

b) Start engine (refer to Para 8.).

c) Allow engine to warm up to operating temperature and idle rpm to stabilize with no load on engine.

d) Ensure fuel control unit (FCU) linkage is in the idle null position by checking that 1/8 to 1/4 inch forward movement of the power lever knob is required to increase rpm (N_G).

Note If unable to achieve an idle null position, recheck power control rigging (refer to 76-10-00, Power Control – Maintenance Practices).

e) Adjust FCU minimum speed adjustment (if necessary) on top of FCU (refer to Figure 2) to give 52 ± 1% N_G. Adjust clockwise to increase rpm (1/16 turn 1% N_G).

CAUTION TURN SCREW IN MINUTE INCREMENTS AS ADJUSTMENT IS EXTREMELY SENSITIVE.

Note If tolerances in Para 10.B.2).e) cannot be achieved refer to Pratt and Whitney Engine Manuals 3021242 Chapter 71-00-00: Fuel Control Unit Checks.

f) Record N_G at idle speed.

3) Adjust prime blade angle.

a) Record outside air temperature (OAT) and pressure altitude (refer to Para 10.B.2).a)).

b) Start engine, refer to Para 8.

c) Allow engine to stabilize.

d) With propeller lever at full INCREASE, advance power lever to obtain 77% propeller speed (N_P).

Note Para 10.B.3).d) should be carried out under zero wind conditions, but an average reading of direct up and down winds can be used. If wind speed is greater than 15 mph, data obtained is unreliable.

e) Read resultant engine torque pressure and record.

f) From curve in Figure 5, read desired torque pressure for ambient conditions.

g) Shutdown engine (refer to Para 8.F).

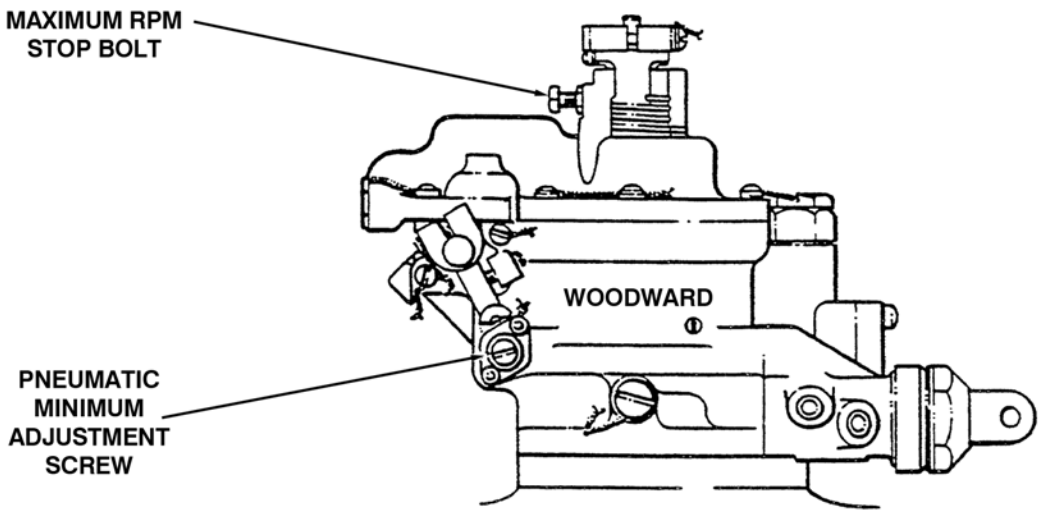
h) Remove propeller spinner.

i) Adjust pitch setting nuts (refer to Figure 3), if necessary, to give desired torque (refer to Para 10.B.3).f)). One flat rotation of nut gives 1 psi torque change.

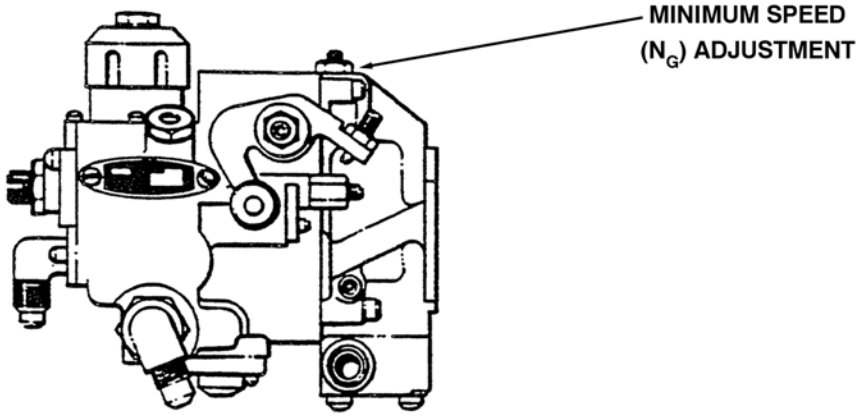
Note Mark position of each nut relative to low pitch stop rod, and turn each nut one flat at a time to ensure nuts are not moved out of position to each other. Adjust nuts out to decrease torque. Minor adjustment only should be required. If any larger change is necessary, the basic setting of beta valve (bottom of clevis flush with valve body) should first be checked (refer to 76-10-00, Power Control – Maintenance Practices, Para 3., Rig Beta Controls).

CAUTION SUBSEQUENT RUNS MAY BE DONE WITH SPINNER REMOVED, BUT IF PROPELLER DE-ICING IS FITTED, SPINNER MUST BE REINSTALLED.

j) Repeat Para 10.B.3).b) to Para 10.B.3).e) and adjust if necessary.

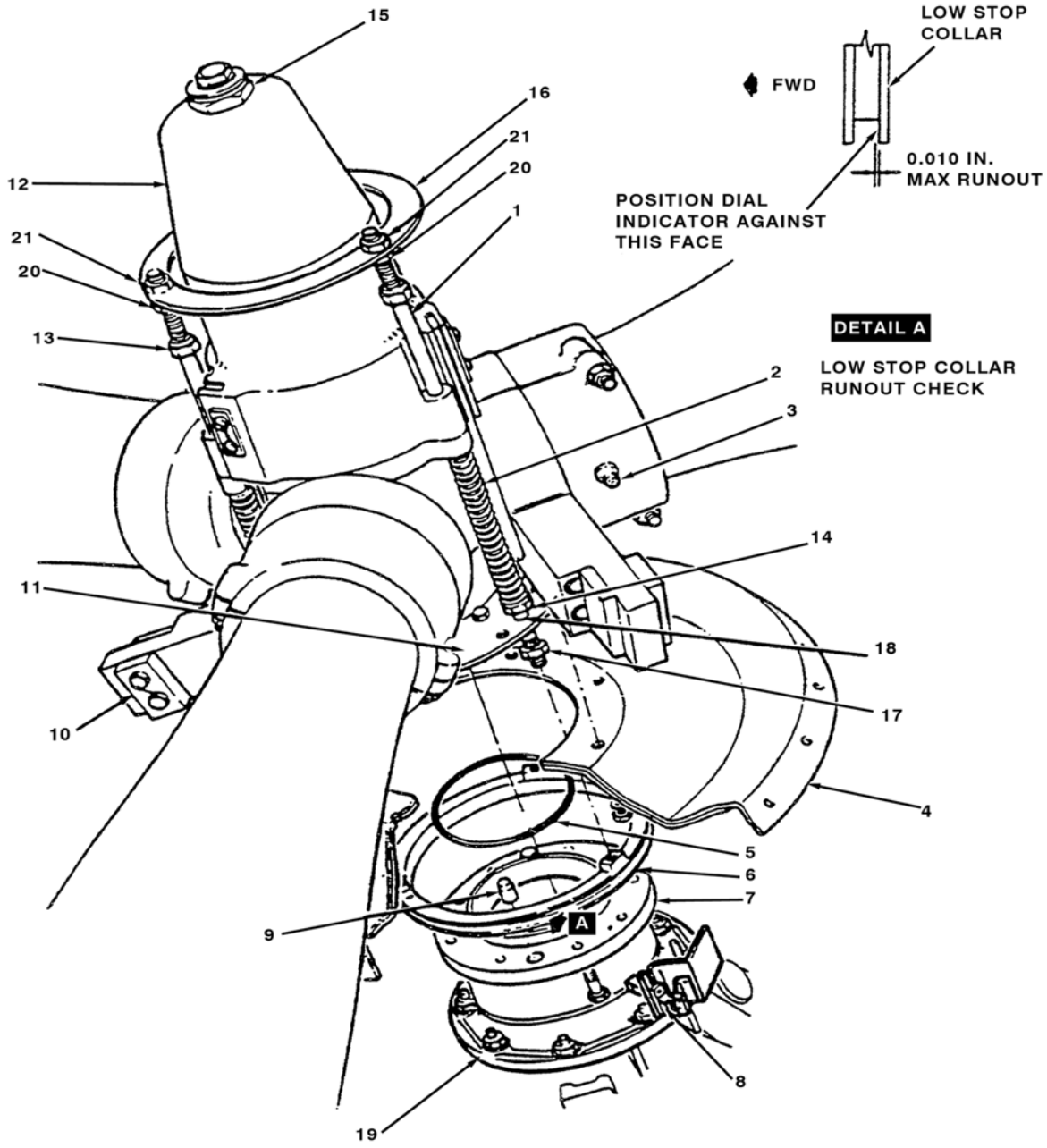


PROPELLER GOVERNOR



FUEL CONTROL UNIT

Running Adjustment Settings
Figure 2



1. LOW PITCH STOP ROD
2. REVERSE RETURN SPRING
3. LUBRICATION FITTING
4. SPINNER BULKHEAD
5. O-RING
6. LOW STOP COLLAR
7. ENGINE SHAFT FLANGE
8. MICROSWITCH SLIDE BLOCK
9. DOWEL PIN
10. COUNTERWEIGHT
11. SPINNER SUPPORT RING
12. PISTON UNIT
13. PITCH SETTING NUT
14. STOP ROD WRENCH FLATS
15. FLEXLOCK NUT
16. ROD END RING
17. LOCKNUT
18. ROD GUIDE
19. ENGINE REDUCTION GEAR CASE
20. ROD END RING REAR LOCKNUTS
21. ROD END RING FORWARD LOCKNUTS

Propeller Installation
Figure 3

4) Adjust idle blade angle (11°).

   a) Start engine (refer to Para 8.), and stabilize at idle with no load on engine.

   b) With power lever at IDLE and propeller lever at full INCREASE, read resultant propeller (N_P) and gas generator (N_G) speeds, and record. Wind effect applies as Note in Para 10.B.3).d).

   c) Shutdown engine (refer to Para 8.F).

   d) From curve in Figure 6, read desired N_P for N_G recorded in Para 10.B.4).b).

      Note If N_P recorded in Para 10.B.4).b) is within tolerance limit (± 1%) of desired N_P shown on curve in Figure 6, disregard Para 10.B.4).e) to Para 10.B.9).

   e) With power lever at IDLE, move propeller lever to minimum rpm position and install rigging pin through propeller and power control pulleys.

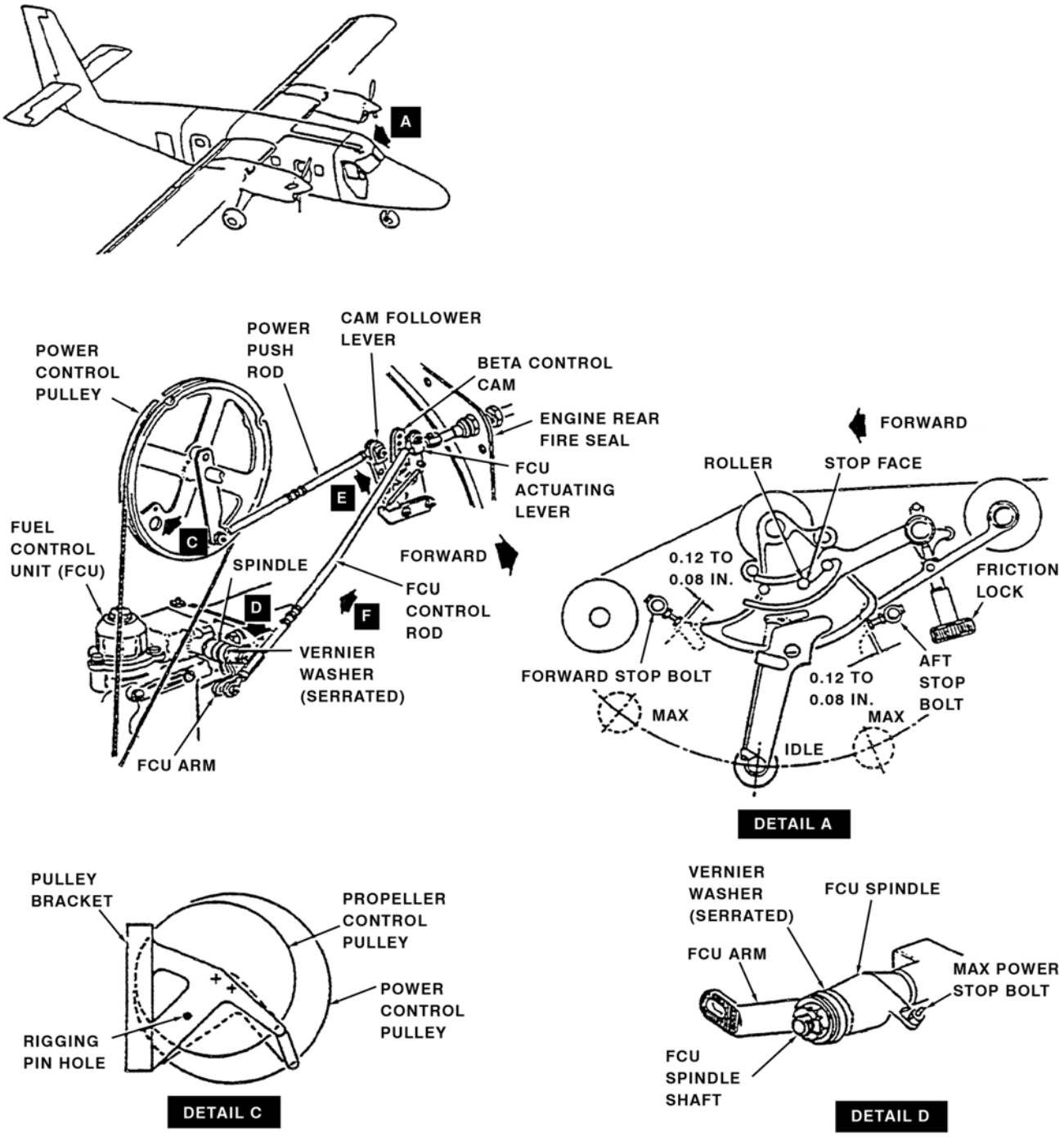
   f) Adjust length of push rod (refer to Figure 4) to bring N_P within limits, shown in Figure 6. Adjust in small increments 3/4 turn of adjuster, is equivalent to 1% N_P. Increasing rod length decreases N_P.

      Note Small adjustment only is permitted. If N_P requires other than minor change, the teleflex control movement in low pitch adjuster should first be checked (refer to 76-10-00, Power Control – Maintenance Practices, Para 3. Rig Beta Controls).

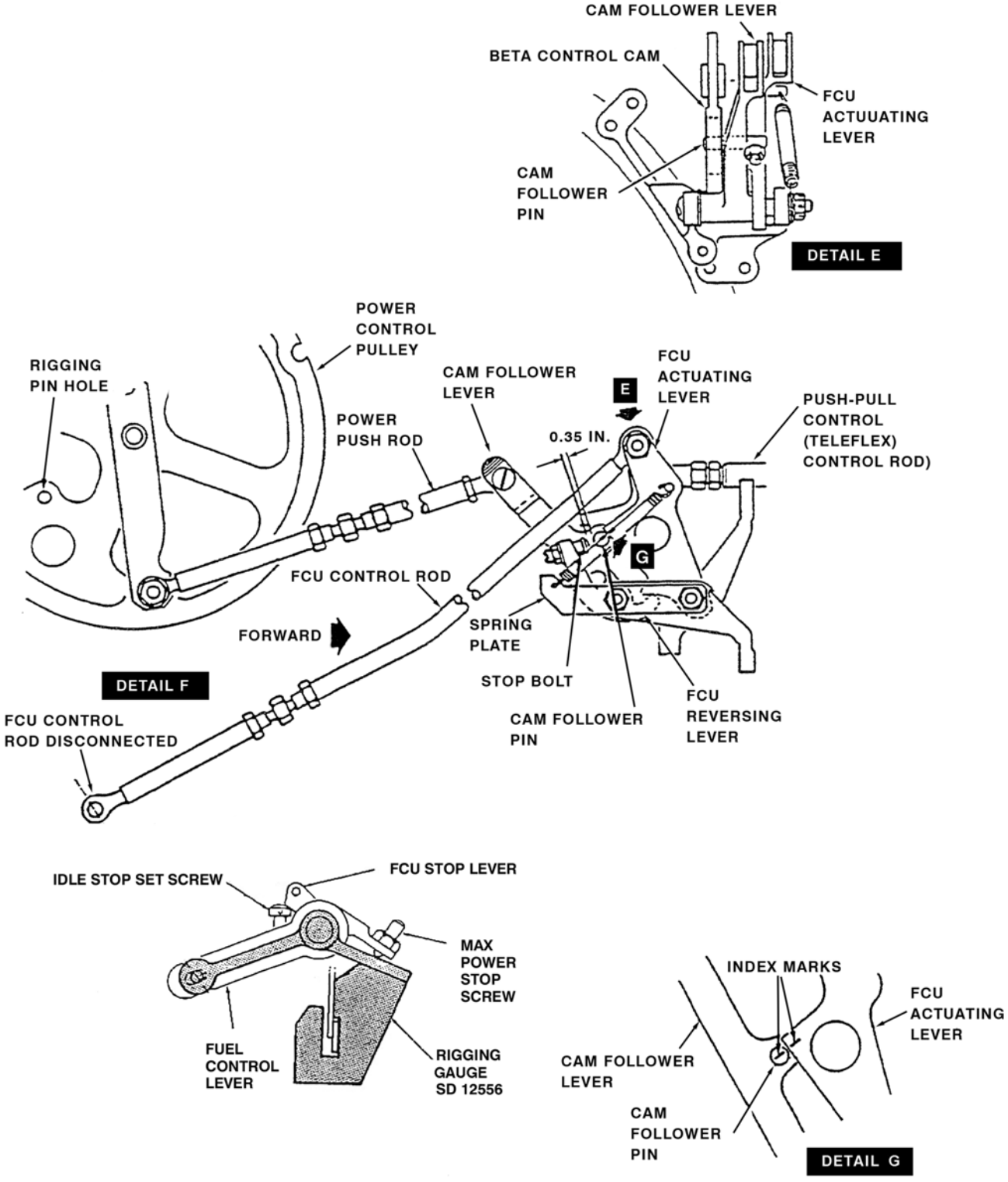
   g) Adjust length of FCU control rod (refer to Figure 4) to compensate for any change in FCU linkage rigging (refer to Rig Power Lever Controls or 76-10-00, Power Control – Maintenance Practices, Para 2. Rig Power Lever Control. Ensure alignment of index marks on cam follower pin and FCU actuating lever.

   h) Remove pulley rigging pin.

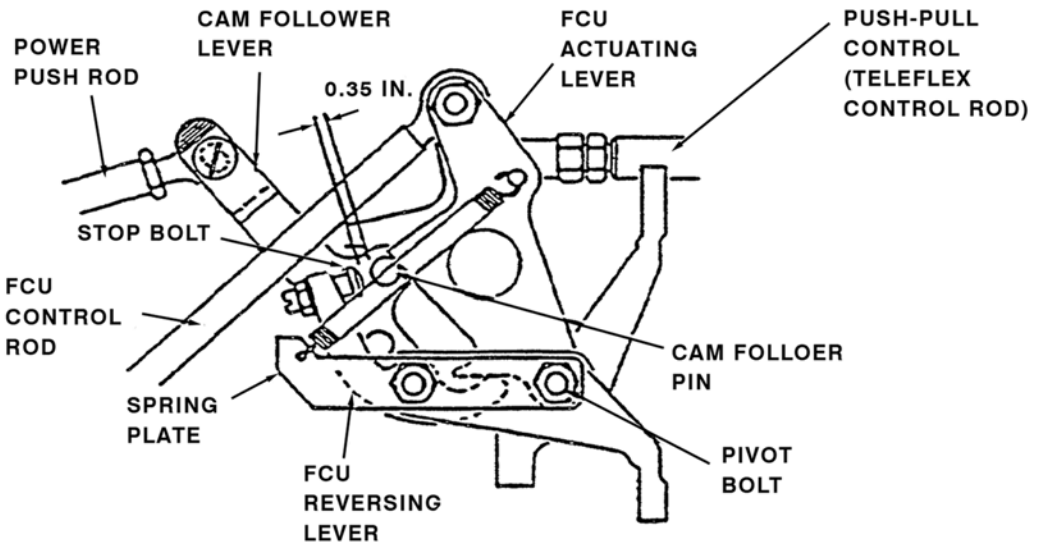
   i) Repeat Para 10.B.4).a) to Para 10.B.4).d).



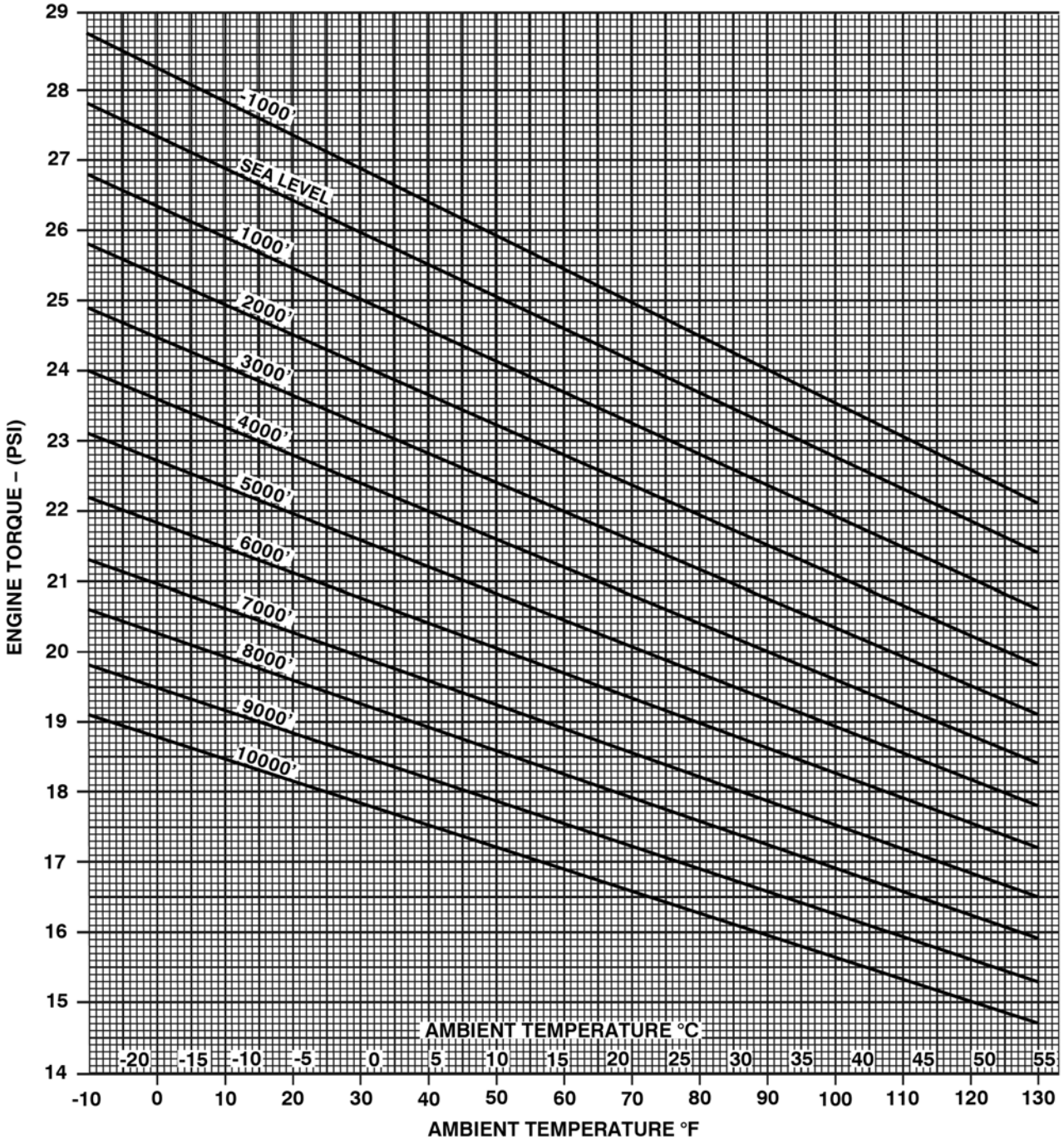
Power Lever Control System Rigging
Figure 4 (Sheet 1 of 3)



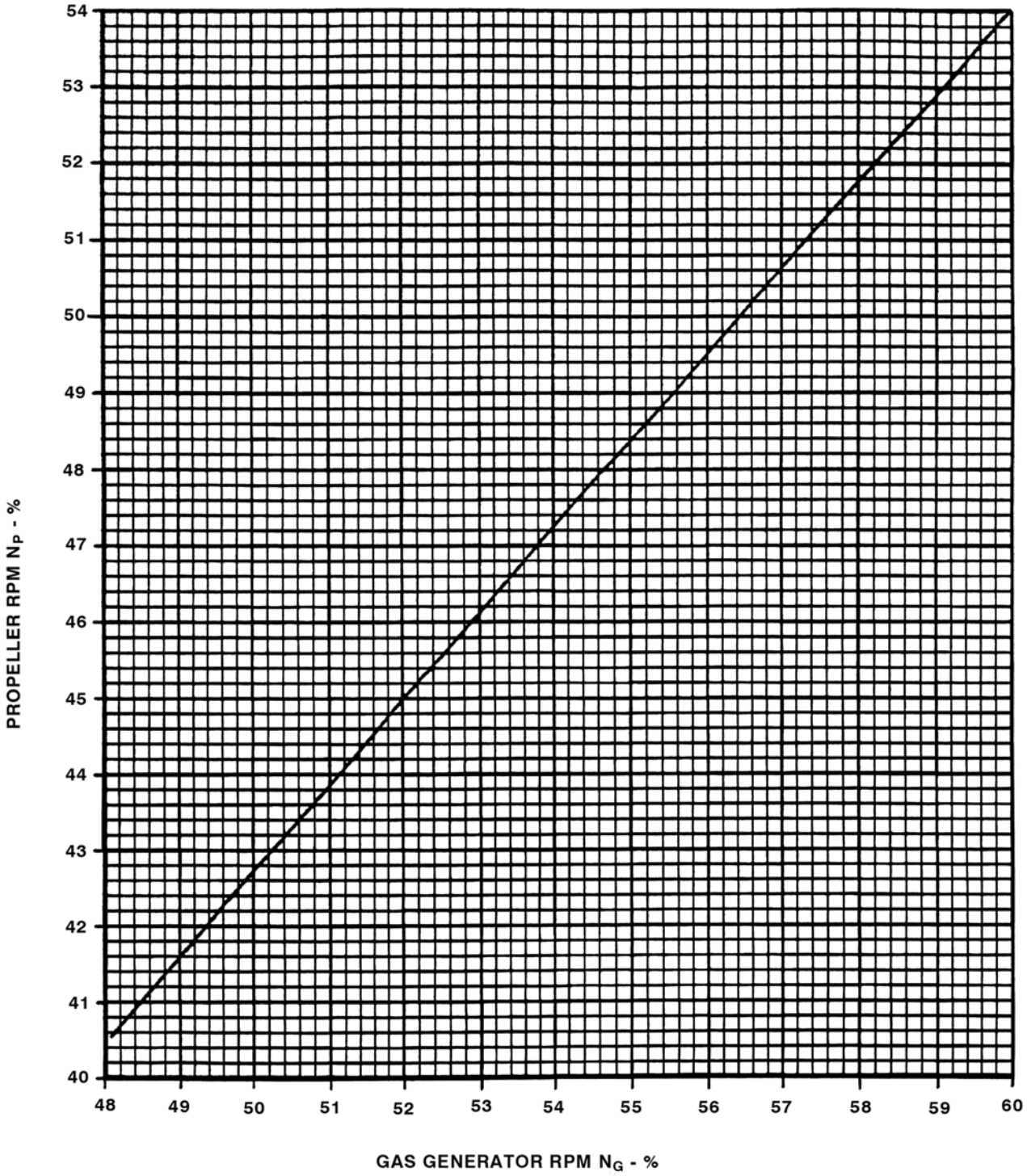
Power Lever Control System Rigging
Figure 4 (Sheet 2 of 3)



Power Lever Control System Rigging
Figure 4 (Sheet 3 of 3)



Prime Blade Angle Curve
Figure 5



Idle Blade Angle Curve
Figure 6

DE HAVILLAND AIRCRAFT OF CANADA LIMITED

TEMPORARY REVISION

PSM: 1-64-2 **AIRCRAFT: DHC-6** **SERIES: 400**

PUBLICATION: MAINTENANCE MANUAL **TEMP. REVISION NO.: TR71-7**

Insert in 71-00-00 after Page 24

POWER PLANT (MOD 6/2012)

REASON FOR ISSUE:

To introduce S.O.O. 6443, Introduction of Optional Power Lever Rigging Procedure.

ACTION:

In Chapter 71, Section 71-00-00 ignore existing pages 25 and 26 and insert revised pages, as shown on Pages 2 to 4 of this Temporary Revision.

-Continued overleaf-

RECORDING INSTRUCTIONS:

Record this TEMPORARY REVISION NO. and the relevant information in the appropriate columns of the TEMPORARY REVISION INDEX provided at the front of the manual.

5) Test beta light system.

Note For test and adjustment of power lever-operated microswitch refer to 61-24-13, Beta Microswitch Assembly (Mod 6/2022) – General Data, Para 3. Adjustment/Test.

Reverse power operation is limited to 50 psi torque pressure with propeller speeds varying between 82% to 87% for ambient temperatures of +125° to –40°F (+51° to –40°C).

a) Run engine at stabilized idle.

b) With propeller lever at full INCREASE, rotate power lever knob and pull power lever approximately 0.5 inch (measured at fairing) into REVERSE. BETA RANGE light should come on after 0.2 inch travel of power lever.

c) Return power lever to IDLE. BETA RANGE light should go out.

d) Repeat steps for other engine.

Note If BETA RANGE lights remain on at idle, this is caused by a slip ring microswitch being incorrectly set, the circuit being closed at idle blade angle instead of approximately 2° below it. To correct this, increase gap (0.030–inch = 1° blade angle) between microswitch roller and slide block (refer to 61-24-13, Beta Microswitch Assembly (Mod 6/2022) – General Data), by adjusting microswitch position until correct operation is obtained.

6) Adjust power lever idle flat.

For aircraft with the basic reverse pickup configuration follow steps Para 10.B.6).a) through Para 10.B.6).d).

a) Run engine at stabilized idle.

b) Rotate power lever knob and pull power lever slowly into REVERSE until gas generator speed starts to increase. Measure movement of power lever on console fairing from IDLE detent to the point of N_G speed increase.

c) If power lever movement (idle flat) measured in Para 10.B.6).b) is less than 1.2 to 1.3 inches, increase 0.350 inch gap between stop bolt and cam follower pin (refer to Figure 4). If power lever movement in Para 10.B.6).b) is greater than 1.2 to 1.3 inches, decrease gap.

d) Repeat Para 10.B.6).a) and Para 10.B.6).b) and, if necessary Para 10.B.6).c), until an idle flat dimension of 1.2 to 1.3 inches is achieved. Ensure that the gas generator pickup point as determined in Para 10.B.6).b) is the same for each power lever.

For aircraft configured with delayed reverse pickup (S.O.O. 6443 required) follow steps Para 10.B.6).e) through Para 10.B.6).h).

e) Run engine at stabilized idle.

f) Rotate power lever knob and pull power lever slowly into REVERSE until gas generator speed starts to increase. Measure movement of power lever on console fairing from IDLE detent to the point of N_G speed increase.

g) If power lever movement (idle flat) measured in Para 10.B.6).f) is less than 1.5 inches, increase 0.440 inch gap between stop bolt and cam follower pin (refer to Figure 4). If power lever movement in Para 10.B.6).f) is greater than 1.6 inches, decrease gap.

h) Repeat Para 10.B.6).e) and Para 10.B.6).f) and, if necessary Para 10.B.6).g) until an idle flat dimension of 1.5 to 1.6 inches is achieved. Ensure that the gas generator pickup point as determined in Para 10.B.6).f) is the same for each power lever.

Note Both engines must be rigged in the same (basic or delayed) reverse pickup configuration, asymmetric rigging is not allowed. Aircraft with S.O.O. 6443 installed may be rigged to the basic reverse pickup configuration if both engines are rigged as such.

7) Adjust propeller control settings.

a) Run engine at stabilized idle.

b) Advance power lever until propeller speed (N_P) is governed at 96% to achieve 620 hp.

c) Move power lever to IDLE.

Note If propeller speed is governed outside limits in Para 10.B.7).b), carry out Para 10.B.7).d) to Para 10.B.7).f), otherwise omit these steps.

d) Shutdown engine (refer to Para 8.F).

e) Adjust maximum rpm stop bolt on propeller governor (refer to Figure 2) to achieve 620 hp propeller speed.

f) Start engine (refer to Para 8.) and stabilize at idle.

g) Retard propeller lever to minimum rpm detent and apply friction lock.

h) Advance power lever to achieve 75% ± 1% N_P. Adjust sliding rod end of teleflex control (refer to Figure 9) if necessary, to achieve 75% ± 1% N_P.

Note Compensating adjustments to match propeller levers can be achieved by moving position of sliding rod end in the slot of speed select lever (refer to Figure 9). Power lever matching can be achieved by moving position of push rod in the slot of cam follower lever (refer to Figure 4).

i) Advance propeller lever to full INCREASE, apply power as required and check propeller speed is 96% + 0% or –1%.

j) Retard power lever to IDLE, and propeller lever to minimum rpm detent.

k) Advance power lever and check 75% ± 1% N_P is achieved.

8) Test maximum setting of fuel governing section of propeller governor (CSU).

a) Position aircraft so that nose wheel is approximately 10° away from engine to be run.

b) Obtain Outside Air Temperature (OAT) and pressure altitude. (Refer to Para 10.B.2).a).

c) Enter take-off power setting curve (refer to Figure 7) with OAT and field barometric pressure or pressure altitude and read off desired torque pressure.

d) Ensure no leaks exist in pneumatic line from CSU to fuel control unit.

e) With engine running at stabilized idle, increase power to desired torque pressure obtained in Para 10.B.8).c). Observe operating limits (refer to Table 1 — Operating Limits).

f) Ensure no hesitation in gas generator speed increase is experienced with advancing power lever and speed remains steady when desired torque pressure is reached. If unable to obtain these conditions, reject propeller governor.

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5) Test beta light system.

Note For test and adjustment of power lever-operated microswitch refer to 61-24-13, Beta Microswitch Assembly (Mod 6/2022) – General Data, Para 3. Adjustment/Test.

Reverse power operation is limited to 50 psi torque pressure with propeller speeds varying between 82% to 87% for ambient temperatures of +125° to –40°F (+51° to –40°C).

a) Run engine at stabilized idle.

b) With propeller lever at full INCREASE, rotate power lever knob and pull power lever approximately 0.5 inch (measured at fairing) into REVERSE. BETA RANGE light should come on after 0.2 inch travel of power lever.

c) Return power lever to IDLE. BETA RANGE light should go out.

d) Repeat steps for other engine.

Note If BETA RANGE lights remain on at idle, this is caused by a slip ring microswitch being incorrectly set, the circuit being closed at idle blade angle instead of approximately 2° below it. To correct this, increase gap (0.030–inch = 1° blade angle) between microswitch roller and slide block (refer to 61-24-13, Beta Microswitch Assembly (Mod 6/2022) – General Data), by adjusting microswitch position until correct operation is obtained.

6) Adjust power lever idle flat.

a) Run engine at stabilized idle.

b) Rotate power lever knob and pull power lever slowly into REVERSE until gas generator speed starts to increase. Measure movement of power lever on console fairing from IDLE detent to point of N_G speed increase.

c) If power lever movement (idle flat) measured in Para 10.B.6).b), is less than 1.2 to 1.3 inches, increase 0.350 inch gap between stop bolt and cam follower pin (refer to Figure 4). If power lever movement in Para 10.B.6).b) is greater than 1.2 to 1.3 inches, decrease gap.

d) Repeat Para 10.B.6).a) and Para 10.B.6).b) and, if necessary Para 10.B.6).c), until an idle flat dimension of 1.2 to 1.3 inches is achieved, to ensure satisfactory zero thrust stop (if fitted) engagement.

7) Adjust propeller control settings.

a) Run engine at stabilized idle.

b) Advance power lever until propeller speed (N_P) is governed at 96% to achieve 620 hp.

c) Move power lever to IDLE.

Note If propeller speed is governed outside limits in Para 10.B.7).b), carry out Para 10.B.7).d) to Para 10.B.7).f), otherwise omit these steps.

d) Shutdown engine (refer to Para 8.F).

e) Adjust maximum rpm stop bolt on propeller governor (refer to Figure 2) to achieve 620 hp propeller speed.

f) Start engine (refer to Para 8.) and stabilize at idle.

g) Retard propeller lever to minimum rpm detent and apply friction lock.

h) Advance power lever to achieve 75% ± 1% N_P. Adjust sliding rod end of teleflex control (refer to Figure 9) if necessary, to achieve 75% ± 1% N_P.

Note Compensating adjustments to match propeller levers can be achieved by moving position of sliding rod end in the slot of speed select lever (refer to Figure 9). Power lever matching can be achieved by moving position of push rod in the slot of cam follower lever (refer to Figure 4).

i) Advance propeller lever to full INCREASE, apply power as required and check propeller speed is 96% + 0% or –1%.

j) Retard power lever to IDLE, and propeller lever to minimum rpm detent.

k) Advance power lever and check 75% ± 1% N_P is achieved.

8) Test maximum setting of fuel governing section of propeller governor (CSU).

a) Position aircraft so that nose wheel is approximately 10° away from engine to be run.

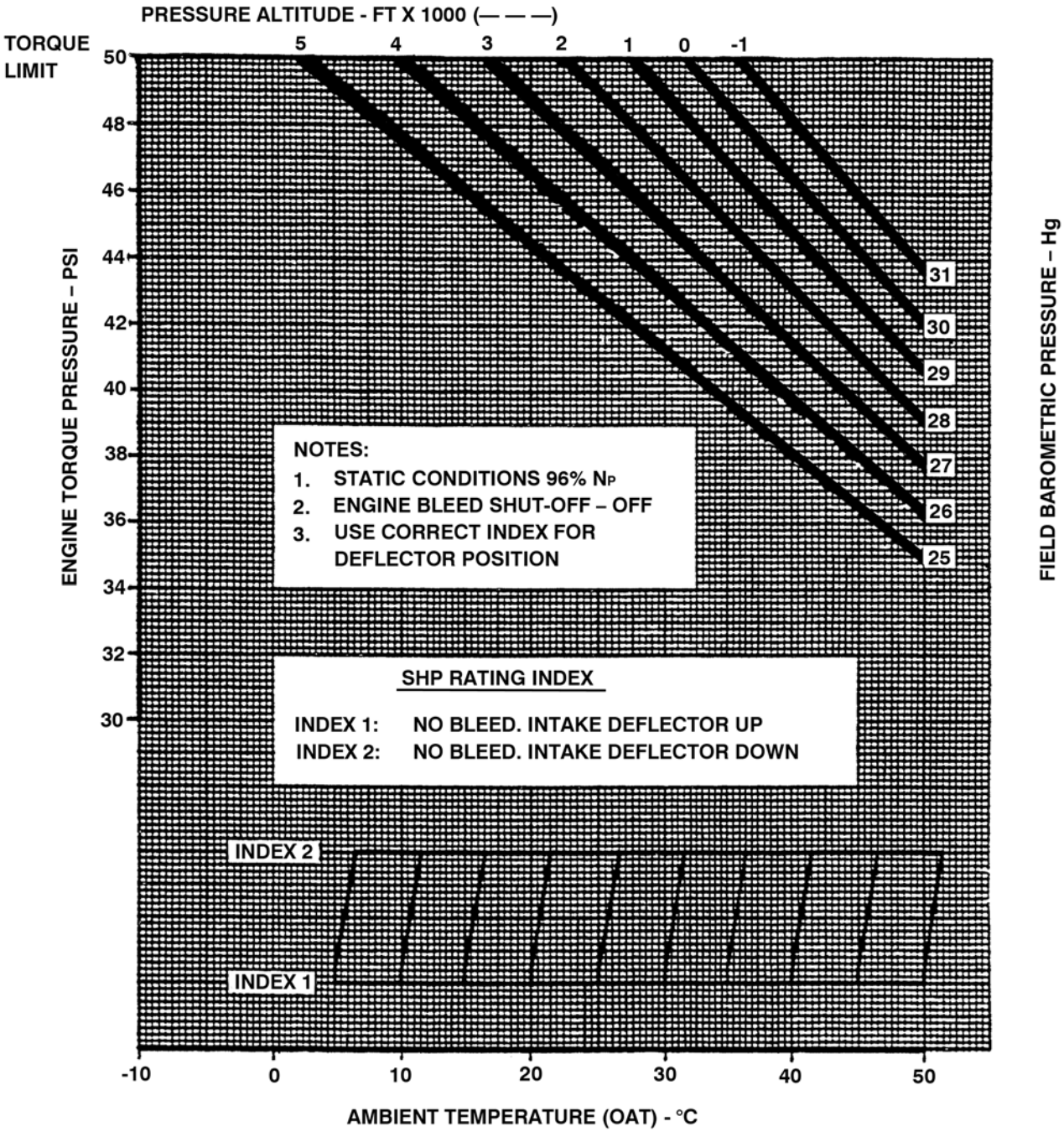
b) Obtain Outside Air Temperature (OAT) and pressure altitude. (Refer to Para 10.B.2).a).

c) Enter take-off power setting curve (refer to Figure 7) with OAT and field barometric pressure or pressure altitude and read off desired torque pressure.

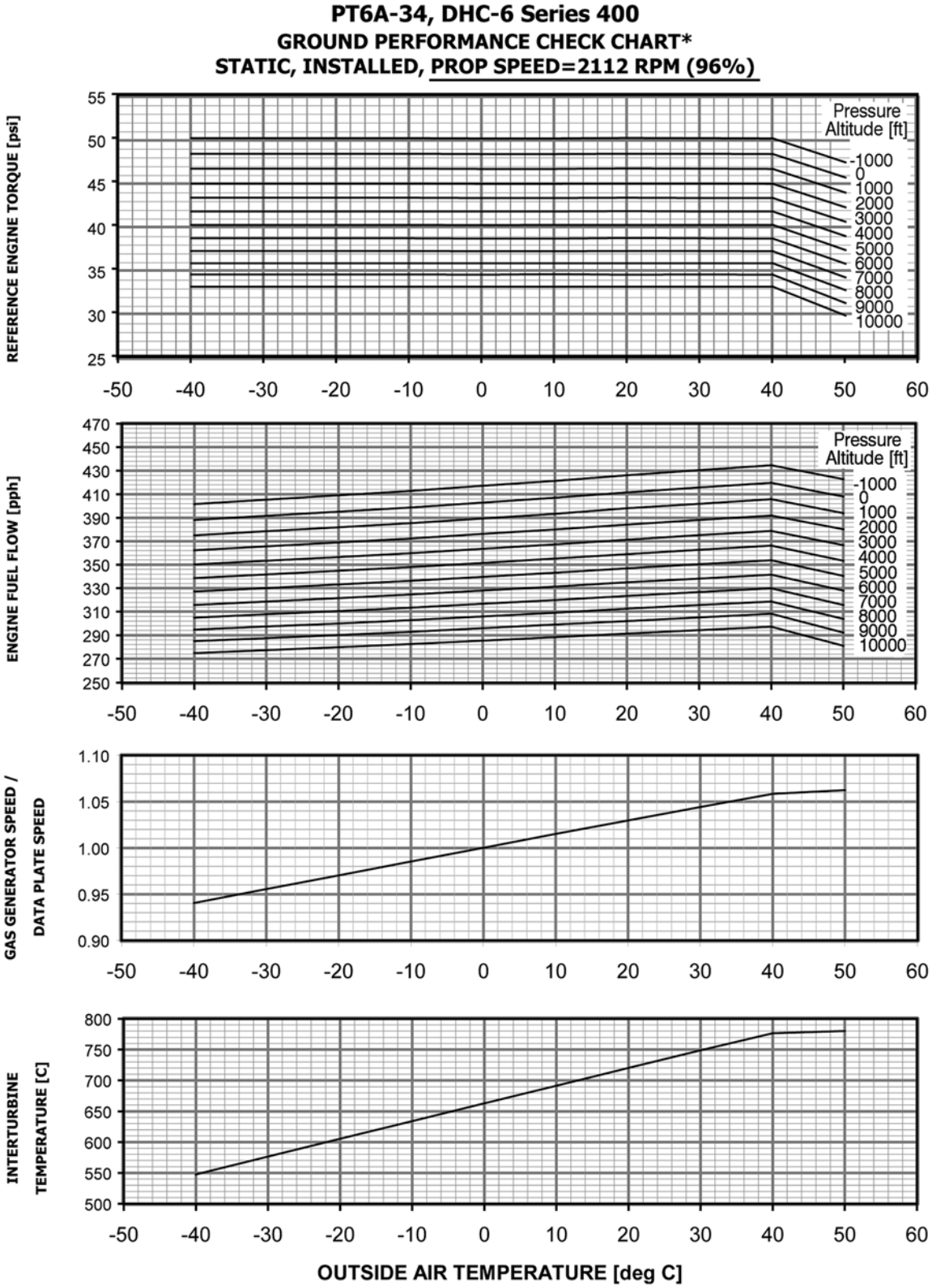
d) Ensure no leaks exist in pneumatic line from CSU to fuel control unit.

e) With engine running at stabilized idle, increase power to desired torque pressure obtained in Para 10.B.8).c). Observe operating limits (refer to Table 1 — Operating Limits).

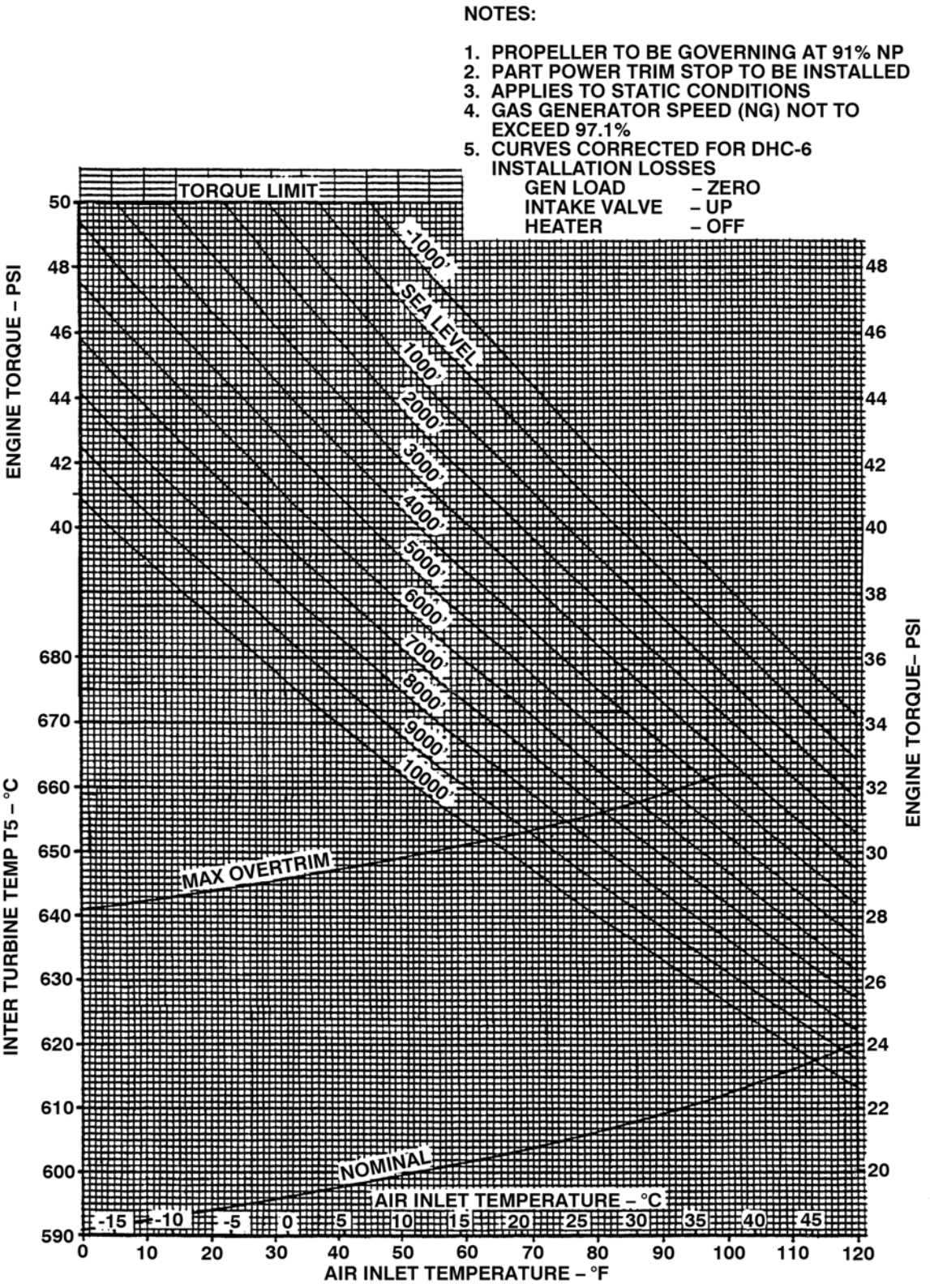
f) Ensure no hesitation in gas generator speed increase is experienced with advancing power lever and speed remains steady when desired torque pressure is reached. If unable to obtain these conditions, reject propeller governor.



Take-off Power Setting Curve
Figure 7



Engine Performance & Data Plate Speed Check
Figure 8



Part Power Trim Check Curve
Figure 9

9) Test minimum setting of fuel governing section of propeller governor (CSU).

   a) Disconnect fuel governor reset arm from fuel governor link.

   b) Move fuel governor reset arm rearwards until internal minimum stop is contacted; secure arm in this position.

   c) Run engine at stabilized idle.

   d) Advance power lever and check that propeller speed (N_P) of 91% + 1% is reached and remains steady.

   e) If N_P is out of tolerance, adjust at pneumatic minimum adjustment screw in 1/16th turn increments (refer to Figure 2).

      Note The adjustment screw connects to an internal eccentric cam, so direction of adjustment rotation cannot be defined for the initial adjustment.

   f) Run engine and recheck N_P is 91% ± 1%. Readjust if necessary.

   g) Shutdown engine (refer to Para 8.F).

   h) Connect fuel governor reset arm to fuel governor link.

10) Test propeller overspeed governor.

   a) Start engine (refer to Para 8.) and stabilize at idle.

   b) Select PROPELLER GOVERNOR test switch to up position.

   c) Advance power lever and observe that propeller governs at 70% ± 2% propeller speed as power is applied.

      Note This should occur at approximately 20 psi, torque pressure varying slightly with changing ambients and altitude. Limit torque pressure is 25 psi during test.

      CAUTION DO NOT RELEASE TEST SWITCH WHILE POWER LEVEL IS ADVANCED.

   d) Reduce power to idle.

   e) Release test switch.

11) Test propeller reset caution light and power/propeller lever interlock.

   a) Run both engines at stabilized idle.

   b) Advance power levers to 75% N_G. Mark position of levers.

   c) Retard propeller levers to just below 90% N_P. Mark position of levers, and check Reset Props caution message is illuminated.

   d) If the caution fails to illuminate, check that the PROPELLER OVERSPEED GOVERNOR circuit breaker is set.

      Note If the caution remains out proceed with Para 10.B.11).e) to Para 10.B.11).l), otherwise omit.

   e) Shutdown engines (refer to Para 8.F) – MASTER switch ON (Pre Mod 6/2314); or EXTERNAL and BATTERY switches ON (Post Mod 6/2314).

   f) Position propeller and power levers as in Para 10.B.11).b) and Para 10.B.11).c).

g) Remove upholstery from cabin roof in vicinity of stations 140.05 to 147.85 (refer to Figure 10).

h) Check microswitches are not actuated.

i) Adjust if necessary by loosening bolt securing slide to cable, and moving slide so that it contacts but does not operate the microswitch.

j) Secure slide to cable by tightening bolt.

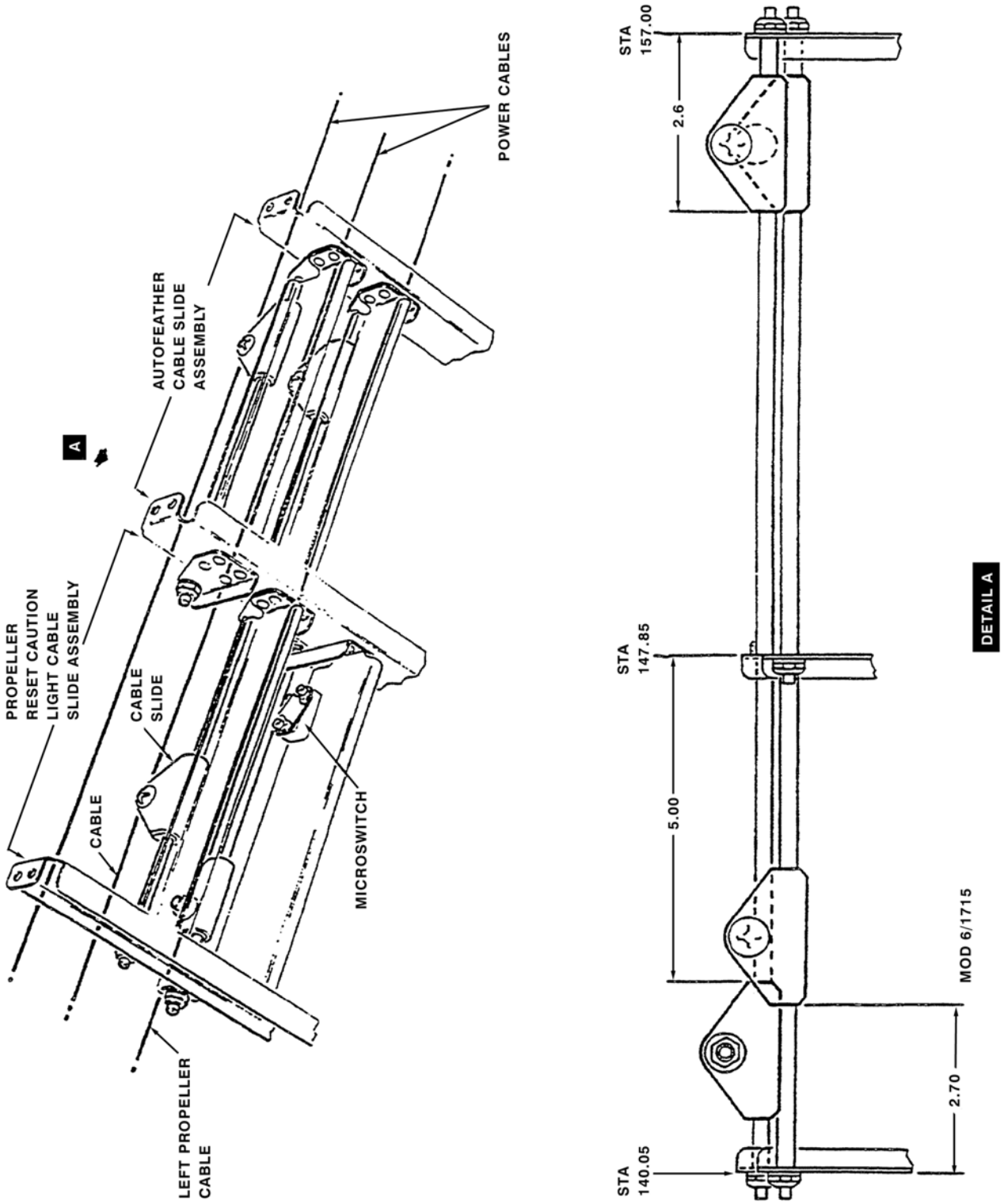
k) Check Reset Props caution message is ON.

l) Replace upholstery in cabin roof.

m) Check the caution goes out when the power levers or propeller levers are advanced past positions marked in Para 10.B.11).b) and Para 10.B.11).c).

n) Position propeller levers to 90% N_P and retard power levers to IDLE.

o) Check that power levers cannot be retarded through the idle gate at the propeller lever setting in Para 10.B.11).n) or at any lower propeller lever setting.



Rigging of Cable Slides
Figure 10

Print Date: 2025-09-18

12) Test autofeather system.

a) Run both engines at stabilized idle.

b) Advance both power levers to obtain torque greater than 26 psi.

c) Select AUTOFEATHER switch to ON and check that SELECTED light illuminates.

d) Hold AUTOFEATHER test switch to test and check that ARMED light illuminates.

e) Retard left power lever to give less than 12 psi torque. Check that ARMED light goes out and two seconds later left propeller feathers. Retard right power lever to give less than 12 psi torque and check that left propeller unfeathers.

f) Release AUTOFEATHER test switch.

g) Repeat Para 10.B.12).b) to Para 10.B.12).f) for right engine, substituting right for left and left for right in Para 10.B.12).e).

h) Advance both power levers to obtain torque greater than 26 psi.

i) Advance left power lever to above 88% N_G. Check that ARM light does not illuminate.

j) Advance right power lever to above 88% N_G and check that ARM light illuminates.

k) Retard left power lever to below 88% N_G and check that ARM light goes out.

l) Retard right power lever to below 88% N_G and check that ARM light remains out.

m) Return both power levers to IDLE. Select AUTOFEATHER switch to OFF and check that SELECTED light goes out.

13) Test engine performance and data plate speed.

Note This test must be carried out with generator OFF, intake deflector retracted and bleed air shut-off valve closed. The Engine Performance and Data Plate Speed Check Curve (refer to Figure 8) is used to establish desired torque pressure, fuel flow, gas generator speed/data plate speed ratio, and turbine temperature. By periodic checking of these parameters, any changes in engine performance can be noted.

a) Obtain ambient air temperature (OAT); maximum limit is 43°C (109.4°F).

b) Obtain pressure altitude (refer to Para 10.B.2).a)).

c) Obtain reference rpm from engine data plate.

d) Enter engine performance check curve with pressure altitude and proceed horizontally left to read torque pressure and record.

e) Enter curve with ambient temperature and proceed vertically upward to intersect appropriate pressure altitude line. From this intersection, proceed horizontally left to read fuel flow in lb/hr and record.

f) Enter curve with ambient temperature and proceed vertically upward to intersect gas generator/data plate speed line. From this intersection, proceed horizontally left to read desired ratio (gas generator speed/data plate speed) and record.

g) Enter curve with ambient temperature and proceed vertically upward to intersect turbine temperature line. From this intersection, proceed horizontally right to read turbine temperature and record.

h) Start engine (refer to Para 8.).

i) Run engine and stabilize at idle.

j) Set propeller control lever to give 96% N_P, with power lever set to give torque setting previously determined in Para 10.B.13).d).

k) Compare observed fuel flow, gas generator speed and turbine temperature with values previously recorded.

l) Ensure that values observed during engine performance and data plate speed test are within the following limits:

	Fuel flow	+15 or –25 lb/hr
Ratio:	Gas generator speed / Data plate ratio	± 0.02
	Turbine temperature T_5	within maximum limit as determined from Figure 8

Note If performance deviates from preceding limits, an instrument or engine fault is indicated (refer to Para 2. and Pratt and Whitney PT6A–34 Maintenance Manual).

m) Shutdown engine (refer to Para 8.F).

14) Check part power trim.

Note Fuel control units are calibrated during bench test to provide maximum power and temperature rating on an 18°C (65°F) day. If engine inlet temperature is other than 18°C (65°F), it is not possible to set maximum gas generator speed stop, since either a turbine temperature (T_5) or a torque limit would be reached first. To overcome this, a part power trim stop is provided. The stop is a movable spacer placed between gas generator maximum power stop anvil and max power stop bolt (refer to Figure 4). The part power trim stop is attached by a screw in its stowed position to the side of the maximum power stop anvil. This check cannot be carried out at ambient temperatures below –1°C (+30°F) as the upper torque pressure limit will be reached before contacting part power trim stop.

a) Loosen screw securing part power trim stop and rotate into position against face of maximum power stop anvil. Tighten screw.

b) Obtain outside air temperature (OAT).

c) Obtain pressure altitude (refer to Para 10.B.2).a)).

d) Enter part power trim check curve (refer to Figure 9) with ambient temperature, and proceed vertically upward to intersect pressure altitude line. From this intersection, proceed horizontally left and read engine output torque. Record value obtained.

e) Enter curve with ambient temperature and proceed vertically upward to intersect turbine temperature line. From this intersection, proceed horizontally left to read turbine temperature T_5. Record value obtained.

f) Start engine (refer to Para 8.) and stabilize at idle with bleed air valves off, generator off and intake deflector retracted.

g) Advance power lever to part power trim stop and allow engine to stabilize at this setting for approximately five minutes at a propeller speed of 91%.

h) Observe torque pressure, turbine temperature and maximum gas generator speed.

CAUTION **GAS GENERATOR SPEED MUST NOT EXCEED 97.1% AT ANY TIME DURING CHECK AND TURBINE TEMPERATURE MUST NOT EXCEED THE VALUE SHOWN ON MAXIMUM OVERTRIM CURVE (REFER TO Figure 9).**

i) Check that the maximum N_G speed is set at 97.1%, the T_5–ITT limit is not exceeded, and the torque meter pressure is equal to or above the referenced pressure (refer to Figure 9). Adjust the N_G maximum power stop bolt on FCU spindle arm, as necessary (refer to Figure 4, Detail D). If the T_5–ITT limit is exceeded, or the torque meter pressure is below the referenced pressure at 97.1% N_G, rectification is necessary prior to further flight.

Note If after adjustment, flight snags indicate power limitations at altitude (operating limit not reached), refer to engine manufacturer for analysis and recommendations.

j) Shut down engine (refer to Para 8.).

k) Stow and lockwire part power trim stop.

CAUTION **PART POWER TRIM STOP MUST BE STOWED AFTER COMPLETION OF PART POWER TRIM CHECK. ENSURE THAT ATTACHING SCREW IS TIGHTENED AND LOCKWIRED.**

l) Lockwire maximum power stop bolt.

15) Match power levers.

a) Attach a strip of masking tape along both edges of face plate between power levers.

b) From Figure 5 obtain required torque pressure for prime blade angle at prevailing ambient temperature and pressure altitude.

c) Start both engines and stabilize at idle.

d) With propeller levers at full INCREASE, advance both power levers to obtain torque pressure determined in Para 10.B.15).b).

e) Mark positions of power levers on masking tape. Shutdown engines and set power levers to IDLE.

f) From marked positions, determine if one power lever is leading.

g) Adjust leading power lever by repositioning power push rod in cam follower lever slot and adjusting its length accordingly (refer to Figure 4). Move power push rod towards inner end of slot to reduce lead.

Note Ensure that cam follower lever position is not moved during adjustment and that power lever is at IDLE.

h) Repeat Para 10.B.15).c) to Para 10.B.15).g) until satisfactory matching is obtained.

Note If unable to match after adjusting as in Para 10.B.15).g), carry out Para 10.B.15).i) as follows:

i) Adjust length of FCU control rod and reposition in slot to FCU arm (refer to Figure 4). Move FCU control rod towards inner end of slot to reduce lead.

Note Ensure that FCU arm and FCU actuating lever are not moved during adjustment.

j) Start both engines and stabilize at idle.

k) Advance both power levers, in increments of 5 psi torque pressure, through full power range, and check matching at each increment.

l) Shudown engines and remove masking tape.

16) Match propeller levers.

a) Attach a strip of masking tape along the outer edges of each propeller slot.

b) Start both engines and stabilize at idle.

c) With propeller levers at full INCREASE, advance both power levers to obtain 95% N_P and tighten friction damper knobs.

d) Retard both propeller levers from full INCREASE to obtain 90% N_P.

e) Mark position of propeller levers on masking tape. Shutdown both engines.

f) From marked positions, determine if one propeller lever is lagging.

g) Adjust lagging propeller lever as follows:

1 Position lagging propeller lever on its 90% mark and secure speed select lever on governor (refer to Figure 11).

2 Disconnect sliding rod end of teleflex control from speed select lever (refer to Figure 11).

3 Position lagging propeller lever in line with 90% mark of leading propeller lever.

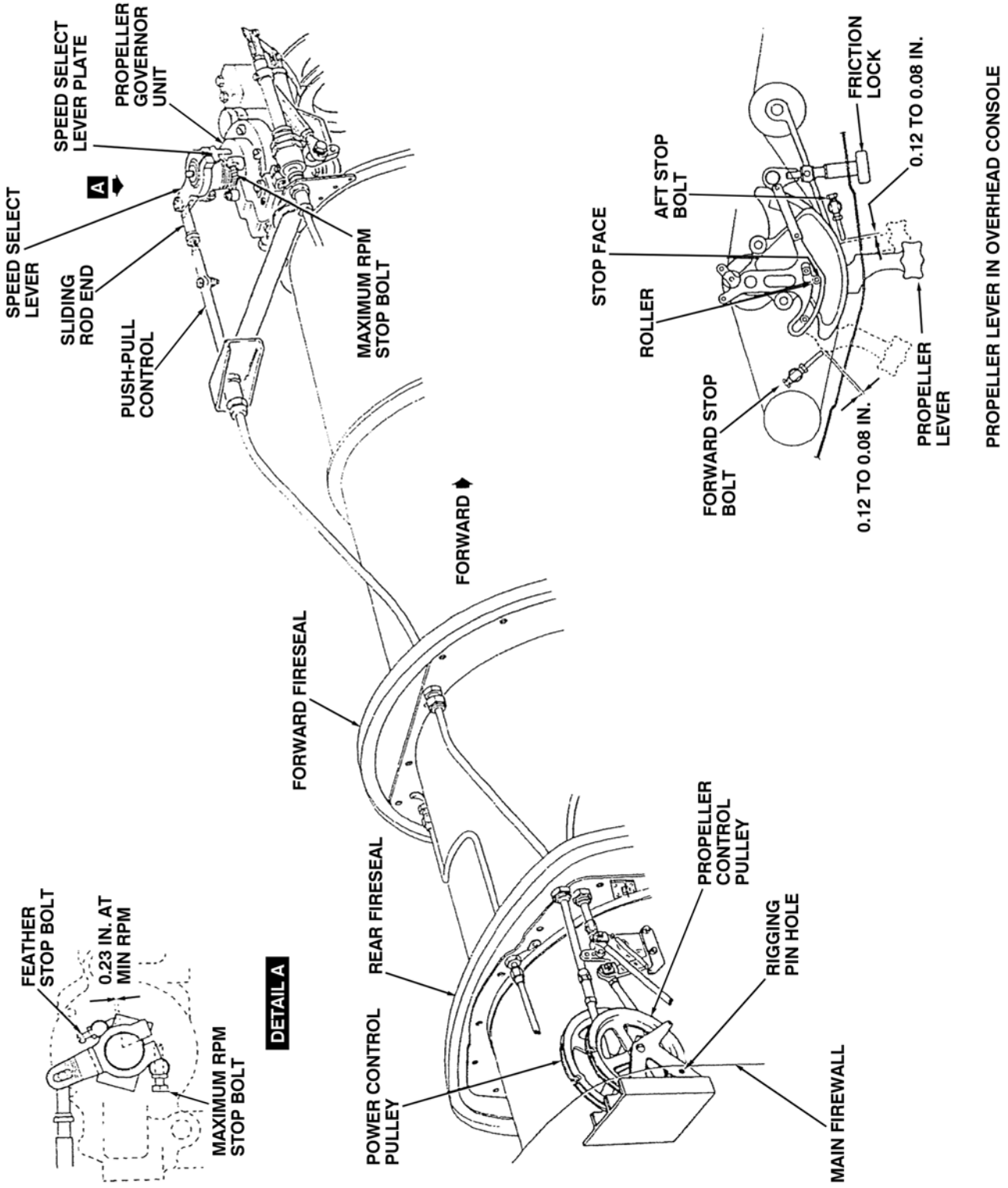
4 Connect teleflex control to speed select lever by moving sliding rod end along slot in speed select lever (refer to Figure 11). Sliding rod end should move towards inner end of slot to reduce lag.

Note If sufficient adjustment is not available in slot, split the difference by also adjusting the leading propeller lever in the opposite direction.

5 Release speed select lever and check that speed select lever plate contacts the maximum rpm stop bolt and feather stop bolt (refer to Figure 11) when the propeller lever is operated through the full range of travel.

h) Repeat Para 10.B.16).b) to Para 10.B.16).g) until satisfactory matching is obtained.

i) Remove masking tape.



Rigging Propeller Control System
Figure 11

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11. Check Ground Running Data

The ground run check sheet (refer to Figure 12) can be used to record information obtained during engine running as detailed in Para 10.B.

GROUND RUN CHECK

DHC-6 Twin Otter (Series 400)

TYPE CHECK		LOCATION	
AIRCRAFT NUMBER		SHUT DOWN	L. R.
ENGINE NUMBERS	L. R.	START UP	L. R.
DATE		RUNNING TIME	L. R.

✲ AMBIENT CONDITION	OAT	FBP	PRESS. ALT

ENGINE RUN	LEFT	RIGHT
MAX. TURB TEMP AT START (LIMIT 1090°C FOR 2 SECS)		

✲ NOTES:
1. FOR FUNCTION OF SYSTEM OR CHECK. USE √ FOR SATISFACTORY OR X FOR UNASTISFACTORY IN SPACE PROVIDED.
2. OBTAIN (a) FROM APPLICABLE CURVE (FIGS 5 TO 9) AND (b) FROM APPLICABLE INDICATOR.
3. ABBREVIATIONS:
   OAT - OUTSIDE AIR TEMPERATURE.
   FBP - FIELD BAROMETRIC PRESSURE.
4. MAKE LOG BOOK ENTRIES AS REQUIRED.
5. TO ACHIEVE 620 HP TAKE-OFF.

✲ GAS GENERATOR IDLE	GAS GEN	RPM (N_G)
52% N_G $^{+1\%}_{-0\%}$	%	%
	%	%

✲ PROPELLER PRIME	TORQUE PRESS.	
BLADE ANGLE CHECK	psi	psi
(77% Np)	psi	psi

✲ PROPELLER IDLE	GAS GEN	RPM (N_G)
BLADE ANGLE (b)	%	%
CHECK	PROP. RPM (N_P)	
	%	%
	%	%

	✲ FUNCTION	
PROP. CONTROL CHECK	PROP. RPM (N_P)	
✲ GOVERNED (96% N_P) (5)	%	%
MIN. RPM (74% - 76% N_P)	%	%
PROPELLER GOVERNOR FUEL SECTION MIN. SETTING CHECK (90% - 92% N_P)	PROP. RPM (N_P)	
	%	%

	PROP. RPM (N_P)	
PROPELLER O/SPEED GOV CHECK (68% - 72% N_P)	%	%

✲ POWER/PROP. LEVER	✲ FUNCTION	
INTERLOCK CHECK		
RESET PROP. CHECK		
AUTOFEATHER CHECK		

✲ DATA PLATE SPEED CHECK (MAX. OAT LIMIT 110°F)	(a)	L.	R.
TORQUE PRESS. psi			
FUEL FLOW pph			
Ng/DATA PLATE SPEED			
TURBINE TEMP °C			

✲ PART POWER TRIM CHECK (MIN. OAT LIMIT 30°F)		L.	R.
TORQUE PRESS. psi			
TURBINE TEMP °C			
MAX. N_G (LIMIT 97.1%) %	X		

CONTROLS MATCH PROP LEVERS

AIRCRAFT SYSTEMS CHECK	
SYSTEM	✲ FUNCTION

REMARKS/FAULTS:

✲ OPERATOR'S SIGNATURE _____

Ground Run Check Sheet
Figure 12

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